



basic education

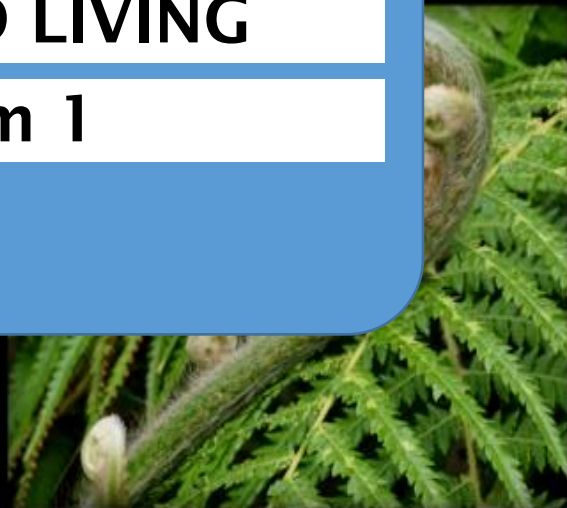
Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATURAL SCIENCES YEAR 3 TERM 1 to 4 TEACHER'S GUIDE



LIFE AND LIVING

Term 1



LIFE AND LIVING

CONTENT OVERVIEW

LIFE AND LIVING	<u>Living and non-living things</u> Living things • <u>Identify</u> and explain • Briefly explain the seven life processes • Non-living • Identify • Explain why things are non-living Practical: Living and Non-living things	<u>Digestive system</u> • Explain concepts • Describe structure • Name the parts and the organs • Describe the features and functions of the parts and organs	<u>Sense Organs</u> Senses • Briefly define concepts • Name and describe animal senses adapted for survival
	<u>Structure of plants</u> • Describe the basic structure of plants • Identify differences between plants	<u>Nutrients in food</u> Food Groups • Briefly define concepts • Categorise food • Describe the functions of food groups	<u>Human skeleton</u> • Briefly define concepts • Function of bones
	<u>Structure of animals</u> • Describe the basic structure of animals • Identify differences between animals	<u>Nutrition</u> • Explain a balanced diet	<u>Human Reproductive System</u> Reproduction • Introduce and explain concepts with examples • Explain the physical changes • Explain the features in the human organisms
	<u>What plants need to grow</u> • Explain requirements to growth • Explain how plants grow from cuttings and seeds	<u>Food processing</u> • Explain concepts, importance of preserving, types of products, methods and benefits	

CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-3	Nutrients in food	Food Groups • Briefly define concepts - Food pyramids - Food groups - Supplements - Fat - Nutrition - Water - Vitamins - Mineral - Carbohydrates - Protein • Categorise food into proteins, carbohydrates, fats, oils, vitamins and minerals. • Describe and discuss the functions and importance of various food groups in the body. Practical activity:
4	Nutrition	• Explain concepts of a balanced diet • Discuss the benefits of a balanced diet • Explain and discuss how malnutrition can lead to diseases Practical: Learners must draw up a healthy menu for breakfast, lunch and supper for one day
5-6	Food processing	Processing Food • Explain concepts: - Processing - Preserve - Artificial - Preservatives - Supplements • Explain the importance and benefits of preserving food. • Explain different methods of food preservation with examples.

7-8	Digestive system	The Digestive System • Explain concepts with examples: - Nutrition - Digestive system - Digestion - Indigestion - Organ of the digestive system • Describe the structure of the digestive system • Name the parts of the digestive system Describe the features and functions of the parts of the digestive system
9 - 10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 50 % Test (Pen and Paper): 50 %		

NUTRIENTS IN FOOD

In this topic we are going to define the following concepts with examples

- Food groups
- Food pyramids
- Carbohydrates
- Proteins
- Fat and oils
- Vitamins and Minerals
- Supplements
- Nutrition
- Water

Key Questions

Why do I have to eat so many different things - why can't I just eat what I like?

Why do we need nutrients in our body?

The teacher must introduce the topic based on the previous knowledge.

Plants and animals need to feed or eat to have energy to carry out the life processes. Animals cannot make their own food and need to eat plants or other animals to get energy. People also have to eat plant or animal products to get energy to grow and develop.

The teacher must explain this to the learners as follows:

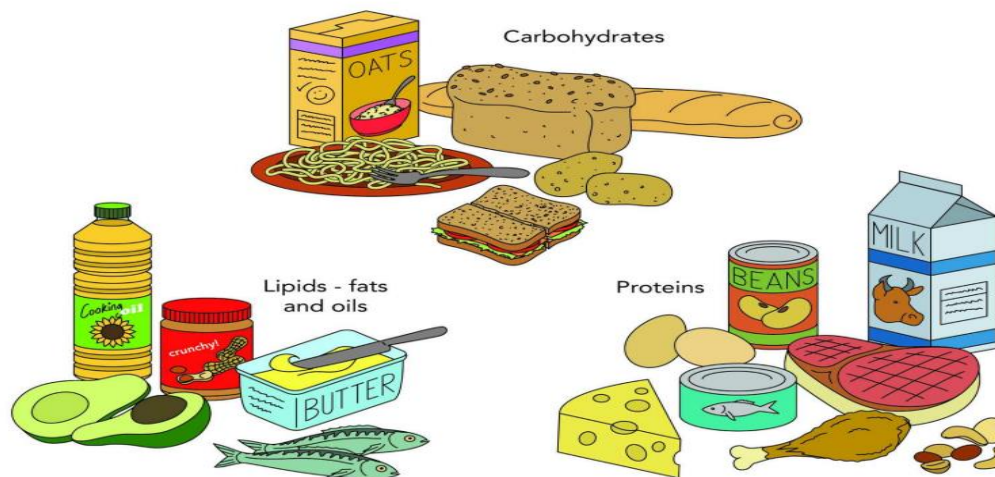
- Go over the **seven life processes** with the learners: feeding, breathing, moving, growing, excreting, sensing things and reproducing.
- Everything animals eat contains nutrients.
- Foods can be grouped according to their functions in the body.
- The four food groups are: carbohydrates, proteins, fats and oils, vitamins and minerals.

Food groups

Classification of food groups

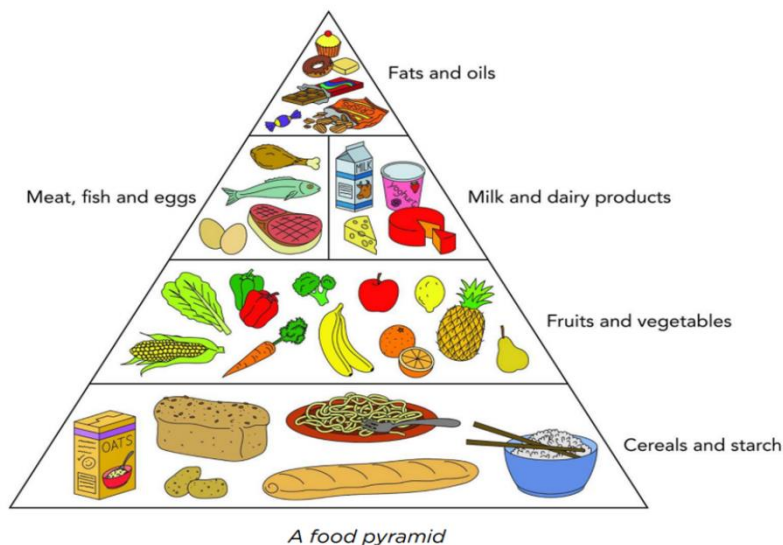
The food we eat can be grouped or classified into different groups. There are a four main food groups and each food group has a different function in the body:

- Carbohydrates
- Proteins
- Fats and oils
- Vitamins and minerals



Examples of the foods in the food groups; carbohydrates, proteins and fats and oils.

Food pyramid gives guide to a healthy balanced diet. It shows how to classify food types into different levels. A balanced healthy diet will include servings from each level of the food pyramid and will show the correct quantities you need to eat. Here is an example of a food pyramid:



We classify food according to these food groups, although most foods contain a mixture of more than one nutrient group.

Let's look at each food group separately and see why each one is important:

Carbohydrates:

- Foods that contain carbohydrates are: pasta, samp, potatoes, cereal, mealie meal, porridge and bread.
- They are the most important source of energy for the body.

- If we eat more carbohydrates than our bodies need, this is stored as fat.



Bread



Rice



Pasta



Potatoes

Proteins

- Protein is found in animal and plant products.
- Animal products are meat, fish, chicken, eggs, milk and cheese.
- Plant products are chickpeas, beans, lentils, nuts and soya.
- Proteins are the building blocks of our bodies - they build the body's muscles.



Meat



Lentils



Almond nuts



Cheese

ACTIVITY: The teacher may ask learners the following questions to check their understanding

1. Why do animals need food?
Animals need food to give their bodies energy to carry out the seven life processes.
2. Why do we need nutrients?
We need nutrients to keep our bodies healthy.

3. What food group do lentils belong to?

Lentils are protein

4. What food group do fish belong to?

Fish are protein

Fats and oils

- Fat is used to protect organs like the kidneys from injury.
- Fat is used to insulate us (keep us warm).
- Foods that contain fats include: butter, margarine, sardines, cooking oils, nuts, peanut butter and avocado pears.



Mixed nuts



Cooking oil



Peanut butter



Margarine

Vitamins and minerals

Vitamin is a food group that helps our bodies need to develop and function normal. They include vitamins A, C, D, E, and K, choline, and the B vitamin. Sources of vitamins are given in the table below.

There are specific foods that contain specific vitamins. The table below shows some important vitamins, their natural sources and/or the food that contains them and some of their functions in the human body.

Name of vitamin	Sources	Function
Vitamin A	Egg yolk, liver, deep-yellow and deep-green fruits and vegetables, fish liver oils	Enables the eye to produce a pigment that helps us to see in dim light.

Vitamin B	Brown rice, whole-wheat bread	They are important in the chemical reactions of breathing.
Vitamin C	Citrus fruit, strawberries, tomatoes, peppers	Strengthens the body's immune system by increasing its ability to fight infections.
Vitamin D	Egg yolk, fish liver oils	Helps the absorption of calcium and phosphate. These nutrients are used for the making of strong bones and teeth.

Mineral is a food group that our bodies need to develop and keeping our bones, muscles, heart, and brain working properly. Many important minerals are contained in the food we eat examples calcium and iron.

Sources of calcium are dairy foods such as milk, cheese and yoghurt. We also get calcium from green leafy vegetables such as kale and broccoli.

Sources of iron are red meat, pork and poultry, seafood, green leafy vegetables such as spinach, dried fruit –raisins and apricot and some cereals, breads and pastas.

Human immune system needs vitamins and minerals to fight diseases.

ACTIVITY: Ask learners the following questions to check their understanding.

1. Give two functions of fats and oils.
Fats and oils keep our bodies warm.
They protect the organs.
2. What system needs vitamins and minerals to help fight disease?
The immune system needs vitamins and minerals to fight disease.
3. Give one function of vitamins and two sources.
Vitamins helps our bodies to develop and function normal
Sources vitamin A –egg yolk, vitamin B – whole wheat bread (**the learner may give any vitamin and source**).
4. Give two types of mineral and their sources.
Calcium – milk, cheese and yoghurt
Iron – spinach, red meat and raisins

PRACTICAL ACTIVITY: Classifying food into the different food groups.

MATERIALS: Collect wrappers and packaging material, like boxes and tins, of the foods you eat and bring these to school.

INSTRUCTIONS:

1. Work in groups of 2 or 3.
2. Choose the 3 foods that you think are the healthiest to eat from the different wrappers and boxes that you have in your group or individual. **(The teacher may decide on individual or group work)**
3. Write a reason next to each of these 3 choices why you think they are really healthy.
4. Sort the different foods into the major food groups.
5. Record your work in the table below.

Carbohydrates	Proteins	Lipids	Vitamins and Minerals

ACTIVITY : Identify different food in a meal according to the food groups

INSTRUCTIONS:

1. Below are pictures of five different meals.
2. Study each meal and classify each food on the plate into one of the food groups (protein, carbohydrates, fats and oils, vitamins and minerals).
3. Remember, some foods may contain more than one food group.
4. In the table, write out each food in the meal, and next to it, write the food group it belongs to:

Meal	Food group
	
	
	
	
	

Supplements

Supplements are products used to improve the diet and often contain *vitamins*, minerals, herbs and meal supplements. It is used to boost the nutritional content of the diet.



Nutrition

Nutrition is the study of nutrients in food, how the body uses them and the relationship between diet, health, and disease. *It* is about eating a healthy and balanced *diet* that is needed to keep growing things.

Water

Water is needed for most body functions as it carries nutrients to all cells in our body. Water allows the body to absorb and digest minerals, vitamins and other substances. Water makes up a majority of our body and is involved in many important functions, including:

- flushing out waste from your body
- controlling body temperature
- helping the brain to function well

Revision:

1. Write the sentence out in full below and fill in the missing words. Foods are classified into the following food types, carbohydrates, **fats and oils**, proteins and **vitamins and minerals**.
2. Name 2 reasons why fats and oils are important in our diet.
Fat is used around organs for protection.
Fat is under the skin to insulate us. (keep our bodies warm).
3. List 3 sources of proteins.
Meat, fish, chicken, eggs, beans, milk, cheese (accept any 3 sources of protein).
4. Choose the correct word from the list below.
body, protein, carbohydrates, starch, repairs
 - 4.1. Milk and cheese are **protein**.
 - 4.2. Bread and potatoes are **carbohydrates**.
 - 4.3. The **body** uses carbohydrates to keep it warm.
 - 4.4. The function of a **protein** is to help our bodies grow.
 - 4.5. Protein also **repairs** our bodies after injury.
5. Complete the table below which shows some vitamins, their natural sources and functions.

Name of vitamin	Sources	Function
Vitamin A		Enables the eye to produce a pigment that helps us to see in dim light
	Brown Rice, whole-wheat bread	There are several types of this vitamin and they have many functions in the chemical reactions of cellular respiration
Vitamin C		Strengthens the body's immune system by increasing its ability to fight infections
Vitamin D	Egg yolks, fish liver oils, can be made in the skin when the ultra-violet rays from the sun convert cholesterol in the skin to vitamin D	

<i>Name of vitamin</i>	<i>Sources</i>	<i>Function</i>
<i>Vitamin A</i>	<i>Egg yolk, liver, deep-yellow and deep-green fruits and vegetables, fish liver oils</i>	<i>Enables the eye to produce a pigment that helps us to see in dim light</i>
<i>Vitamin B</i>	<i>Brown Rice, whole-wheat bread</i>	<i>There are several types of this vitamin and they have many functions in the chemical reactions of cellular respiration</i>
<i>Vitamin C</i>	<i>Citrus fruit, strawberries, tomatoes, peppers</i>	<i>Strengthens the body's immune system by increasing its ability to fight infections</i>
<i>Vitamin D</i>	<i>Egg yolks, fish liver oils, can be made in the skin when the ultra-violet rays from the sun convert cholesterol in the skin to vitamin D</i>	<i>Used for the making of strong bones and teeth</i>

6. Briefly explain why water is important for our bodies.

Water is important for most body functions as it carries nutrients to all cells in our body.

Water allows the body to absorb and digest minerals, vitamins and other substances.

Water makes up a majority of our body and is involved in many important functions, including:

- flushing out waste from your body
- controlling body temperature
- helping the brain to function well

7. What is the function of minerals to our bodies?

Minerals help our bodies to develop and keep our bones, muscles, heart, and brain working properly.

TOPIC 2: NUTRITION

In this chapter we are going to look more closely at the food we eat to see why certain foods are considered healthy and others unhealthy. We will then investigate how the food from our plates gets to our bodies and why our digestive system is so well adapted for its job.

Nutrition refers to the food needed to keep growing things healthy

A diet is the kinds of food a person or animal eat.

What is a balanced diet?

A balanced diet contains the **correct quantities** of nutrients, from the basic food groups, that a person needs each day for healthy growth and activity.

Our human bodies are very active. Our bodies need a huge variety of different nutrients and substances in order to perform all these processes. We obtain these nutrients from the food we eat. The human body needs a balanced, healthy diet to keep functioning properly.

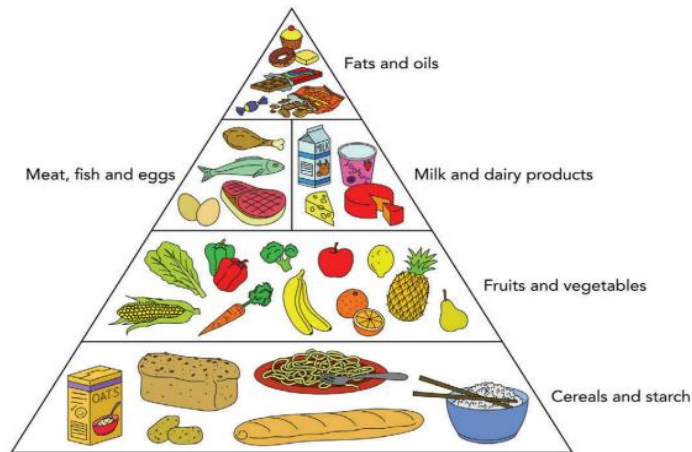
QUESTION

Discuss the word diet with your class and come up with a definition for this word. Write it below.

For nutritional scientists (people who study the effects of diet on health and well-being), the word diet means the sort of food that people regularly eat. When we say

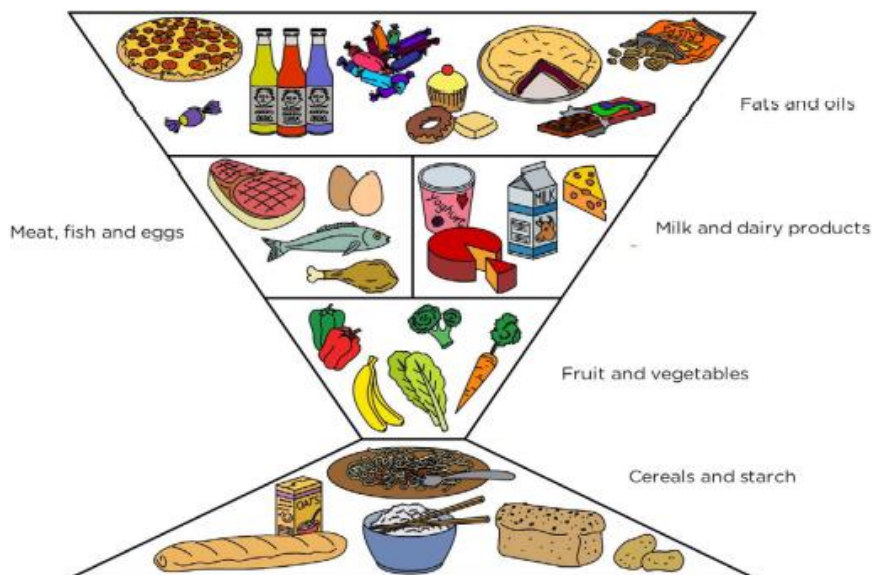
we eat a healthy balanced diet we mean that we eat food that contains adequate or the right amounts of all the nutrients our body needs to function

In the previous topic we dealt with the food pyramid that provides a guide to a healthy balanced diet. It shows how to classify food types into different levels. A balanced healthy diet will include servings from each level of the food pyramid and will show the correct quantities you need to eat.



A food pyramid.

Some people eat a very unhealthy diet and turn the food pyramid upside down! They eat far more sweets, cakes, fatty rich food and far less vegetables and starches!



The food pyramid of an unhealthy person's diet.

Let's take a closer look at what makes up a healthy diet.

The foods that we eat can be divided into different groups:

- proteins
- carbohydrates
- fats and oils
- vitamins and minerals

Different cultures and religions follow different diets. Some cultures will only eat certain types of food and will avoid other combinations. Some religions might restrict their followers to only certain foods while others have no real dietary laws. Within South Africa, we have a very diverse population with people from many cultures, backgrounds and religions. This makes our country a truly unique, diverse and interesting place in which to live!

ACTIVITY:

Evaluating a diet to see if it contains all the food groups.

MATERIALS:

- A record of the weekly diet of Farrah's brother, Rajesh

INSTRUCTIONS:

1. Carefully study Rajesh's weekly diet.
2. Answer the questions that follow.

	Breakfast	Lunch	Supper
Monday	toast with peanut butter, coffee with milk and 3 teaspoons of sugar	two-minute noodles, cream soda, jelly (while watching TV)	spaghetti bolognese (with chopped carrots in the mince) with cheese sprinkled on top, fruit juice, tea with milk and 2 teaspoons of sugar
Tuesday	toast with peanut butter, coffee with milk and 3	two-minute noodles, Coke, toast with peanut butter, popcorn	Mom and Dad worked late - scrambled egg on toast with baked beans; coffee with

	teaspoons of sugar	(while watching TV)	milk and 3 teaspoons of sugar
Wednesday	Late for school - didn't eat.	2 eggs and toast with cheese, tea with milk and 3 sugars, more toast with jam and peanut butter later (while watching TV)	Hawaiian pizza, coke, cream soda, chocolate milkshake
Thursday	toast with peanut butter, coffee with milk and 3 teaspoons of sugars	indoor hockey practice after school - pie from tuckshop and Coke, chocolate and packet of hot chips	roast chicken, butternut, baked potatoes, salad, fruit juice, coffee with milk and 1 teaspoon of sugar
Friday	toast with butter (peanut butter is finished), coffee with milk and 3 teaspoons of sugar	band practice after school - pie from tuckshop and coke, sweets	takeaway chicken in a bucket, hot chips, Coke, tea with milk and 3 teaspoons of sugar, popcorn

QUESTIONS:

- Let's use our knowledge of a food pyramid to see how healthy Rajesh's diet for the week was. Use the tallying method to keep score of everything he ate: to tally up results you make one line down for every item you count and when you get to the fifth item you draw a diagonal line through the other four. That makes it easy to count in fives and count the few left over at the end

Total Fats and Oils	Total Proteins	Total Vitamins & Minerals	Total Carbohydrates

Teacher's Note

Caution learners to use small tallies else they might run out of space. This activity is not meant to be scientifically accurate. However, if they tally each time a food is mentioned and for sugar maybe 1 tally for sugar.

2. List the times and days when Rajesh ate fruit or vegetables.
Monday, Rajesh had carrots in the spaghetti bolognese and on Thursday he had salad and butternut with his meal.
3. What important food group is Rajesh missing from his diet if he does not eat enough fruit and vegetables?
He will not be getting enough vitamins and minerals.
4. Can you predict what will happen if Rajesh continues with this diet?
If he is not already very overweight he soon will be overweight and will develop problems that accompanies this - diabetes, bad skin, aching joints, etc.
5. Now that you have tallied up the different food groups that Rajesh ate in a week, do you think he ate a balanced diet? Why do you say so?
Rajesh's diet was not balanced because Rajesh did not eat the correct quantities of food from the different food groups and therefore I don't think the diet is very balanced.
6. Now that you have completed this activity, explain why different portions of the different food groups are necessary for a balanced diet?
A balanced diet contains sufficient quantities of food from all four food groups, as well as water and fibre to stay healthy.

Benefits of eating a healthy diet

Our human bodies are very active. Our bodies need a huge variety of different nutrients and substances in order to perform all these processes. We obtain these nutrients from the food we eat. The human body needs a balanced, healthy diet to keep functioning properly.

Food Group	Benefits
Proteins	Proteins are our bodies' building blocks. They build and repair body cells and tissues. Foods rich in protein are: fish, meat, poultry, eggs, cheese and other food from animal sources. There are also many sources of protein from plants. For example: products made from soya beans, peas and beans, nuts and seeds.
Carbohydrates	Carbohydrates are the main supply of energy for our bodies. Examples of foods that contain carbohydrates are: whole grain bread, potatoes, pasta, rice, fruit, vegetables, maize and legumes. Unfortunately many

	people eat too many carbohydrates, especially processed carbohydrates like sweets and biscuits, chips, pastries, soft drinks and sweetened fruit juice
Fats and oils	Fats and oils are important for many body processes: Fat protects and insulates your organs
Vitamins	Vitamin A for example, helps your immune system to fight infections, and keeps your skin healthy. Vitamin D helps to regulate the amount of calcium and phosphate in the body, important for bone, teeth and muscle health. Vitamins also support normal growth and development.
Minerals	Minerals are necessary for three main reasons, building strong bones and teeth, controlling body fluids inside the body, and turning the food you eat into energy.

Water not a food group but is essential for the body: Water is necessary to help our blood carry nutrients and waste around the body and to help the chemical reactions that occur in our bodies. Water forms most of sweat, saliva and tears

Malnutrition can lead to diseases

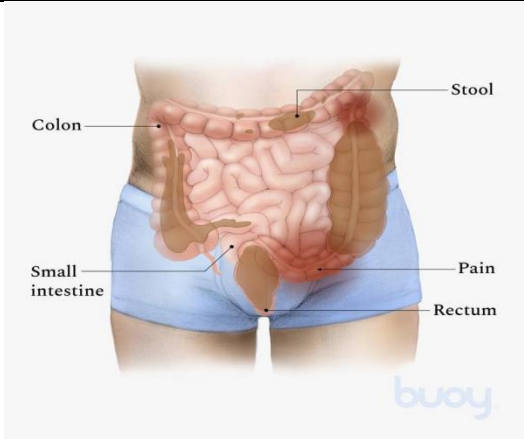
Some diseases and conditions

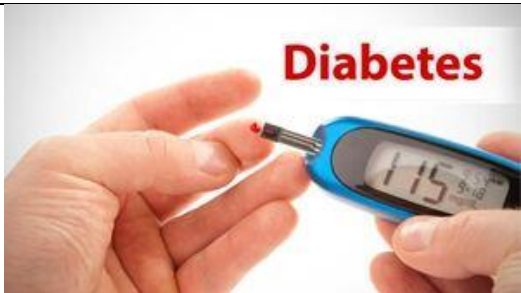
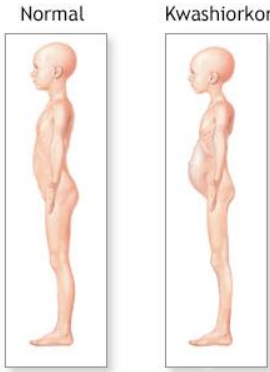
Teacher's Note

It is not necessary for learners to know the difference between a disease and a condition, but generally a "disease" refers to an illness or some deviation from the normal functioning of the body or organ. A "condition", however, is a broader term and often refers to the state of something. A disease is a deviation in the normal functioning of the body which is indicated by certain symptoms.

Malnutrition is caused by a **lack of nutrients**, either as a result of a poor diet or problems absorbing nutrients from food.

Often we get diseases and conditions because our bodies do not get the necessary nourishment. *Some of these diseases/ conditions are:*

	Description	Illustration/picture
Tooth decay:	When you eat or drink too much sugar, the bacteria in your mouth makes acids that eat away at the tooth surface (called enamel), and causes cavities to form.	 Too many sweets can cause cavities to form in your teeth. ¹ Remember to brush your teeth to help prevent tooth decay! ²
Rickets:	This is a condition mostly seen in babies and children. When children do not get enough fruit and vegetables and do not get enough Vitamin D, their bodies cannot absorb calcium and they cannot grow strong bones.	 Vitamin D deficiency rickets: a problem for our times – Corpus
Constipation:	This is a painful condition (not a disease) that occurs when bowel movements become difficult and take place days or even weeks apart. An unhealthy diet is often to blame, but there are other causes too. The best way to treat constipation is to eat a healthy diet with lots of fibre in the carbohydrates, to exercise regularly and	 constipation - Google Search

	to drink plenty of water.	
Diabetes:	This is a disease that affects the body's ability to use blood sugar for energy. Diabetes symptoms may include an increased thirst and urination, blurred vision and excessive tiredness.	 Diabetes John Peel Wiki Fandom
Anorexia nervosa:	This is life-threatening eating disorder where the patient starves themselves and deprives their bodies of nourishment to cause themselves to lose a large amount of weight.	 Anorexia nervosa - Simple English Wikipedia, the free encyclopedia
Kwashiorkor:	This is a disease that affects many children. It is a lack of protein and iodine in their diet. Children suffering	 Kwashiorkor Information Mount Sinai - New York

ACTIVITY

1. What do you understand by the term "balanced diet"? Write a description below.

It refers to the amounts of each food group we need to eat regularly to stay healthy/ A healthy diet provides all the nutrients, such as carbohydrates,

proteins, fats, vitamins, minerals and fibre that are needed for well-being, and contains them in correct amounts/ proportions.

2. Draw a food pyramid of the food that you eat and assess whether it has the correct shape of a balanced diet.

(Mark individual learner's food pyramids based on the one used earlier in the book.)

3. Rajesh thinks that he can get enough water in his diet by drinking Coke and coffee with sugar in it. Explain to him if this is true or not and why.

Not true. Coke contains too much sugar and coffee is diuretic (causes dehydration)

4. Write a paragraph where you explain some of the possible causes of a poor diet.

Learners should discuss some of the causes identified in the chapter, and any others they can think of, including:

- Diseases and infections
- Dietary practises
- Poverty and high food prices
- Food shortage

TOPIC 3: FOOD PROCESSING

NOTE TO TEACHERS

Start this topic by taking learners back in history of food processing: Why the need to preserve food? How the different cultures used these methods?

Many indigenous South African cultures leave milk to thicken in a hide sack or calabash to form a favourite semi-solid called amasi.

The Ndebele people of Southern Africa eat a plant called Umrhorho. It is cooked or dried by spreading it out in the sun so that it can be stored for the cold winter months.

The Swazi people make Marula beer by placing the ripe fruit in large traditional pots and letting them ferment.

Biltong was invented by the Voortrekkers who travelled by ox-wagon during the Great Trek. They needed to preserve meat for the long journey. The raw meat was preserved using salt, vinegar and spices.

People have been processing food for thousands of years. Before the invention of fridges and freezers, people had to collect food in summer and store it because in winter food is often scarce. They found out that they could preserve food like meat and fish by salting and spicing, and then drying it - this is called **curing** the meat. In South Africa **biltong** and **bokomos** are examples of cured meat and fish.



Bokkoms (cured fish) ¹



Pieces of meat that are hung to dry

WHAT IS FOOD PROCESSING

Food processing is a process of changing raw animal or plant material into products and food that we can eat and use.

Food processing can include: cooking, baking, roasting, grilling, smoking, drying, spray drying, juice concentrating, freezing and milling (wheat).

We process food to make it more edible (preparing / cooking), last longer (preserving) and to increase the nutritional value (fermenting).

ACTIVITY: Describing processed foods

INSTRUCTIONS:

1. Look at the following photographs of various foods.
2. Each one has either been processed in some way, or needs to be processed before we can eat it.
3. Briefly describe how each food has or will be processed.

FOOD	HOW EACH FOOD HAS OR WILL BE PROCESSED
 <i>Raw meat</i>	Cooking, drying, freezing
 <i>Yoghurt</i>	Fermentation
 <i>Pickled gherkins</i>	Pickling

 <i>Glacé cherries</i>	Adding sugar	
 <i>Fried eggs</i>	Cooking	
 <i>Butternut</i>	Cooking and freezing	
 <i>Frozen foods</i>	Freezing	

PRESERVING FOOD

Preserving is a process of making food last longer.

The following are methods in which food is preserved:

- Sun drying
- Fermenting dairy into cheese, buttermilk (Amasi) or yogurt, or barley or grapes

- Pickling vegetables (onions or gherkins, for example)
- Salting and spicing meat to preserve it (then drying it)
- Adding sugar to berries and/or fruit to make preserves
- Pasteurisation: using just the right amount of heat to warm up milk or juice for example, to extend the shelf life.
- Cooking: roasting, smoking, baking, frying
- Toasting
- Freezing or refrigerating
- Freeze drying
- Spray drying
- Making juice concentrates

ACTIVITY: IDENTIFY WAYS OF PRESERVING FOOD

1. Copy and complete the table below.

PROCESSED FOOD	MAIN METHOD OF PRESERVING	EXTRA EXAMPLES
Biltong 	Drying or salting	Dried fish
Dried tomatoes 	Drying	Mushrooms
Frozen peas 	Freezing	Frozen Meat

ACTIVITY: METHODS OF PRESERVING

Next to each item, write down the type of preservation it might have gone through and the reasons for that type of processing.

FOOD ITEM	TYPE OF PRESERVATION	REASON FOR PROCESSING
Canned fruit  https://www.tasteofhome.com/article/canned-fruit/	<i>Canning/ Adding sugar</i>	<i>Make it last longer</i>
Amarhewu  https://www.mzansistylecuisine.co.za/strawberry-mango-mageu/	<i>Fermenting</i>	<i>Add nutritional value, make it last longer</i>
Dried Fruit  https://heidisdriedfruitandnuts.co.za/product/mixed-fruit-salad-choice-grade-1kg/	<i>Sun drying</i>	<i>Make it last longer</i>
Frozen chicken pieces	<i>Freezing</i>	<i>Make it last longer</i>



<https://www.southernliving.com/food/meat/chicken/cooking-frozen-chicken>

ACTIVITY: Food processing methods

Food Processing Project Now that you have a better idea of the different ways that raw food can be processed, choose one particular method that you found really interesting to further research and then process the food at the end.

Teacher's Note After researching the processing method, learners are encouraged to actually carry out the food processing, either at home or in class. On a specified day when learners present their projects, they can bring the actual food product to class.

INSTRUCTIONS:

1. You need to read as much as possible about your chosen method of food processing.
2. You will need to interview at least one person who uses this method of food processing. Before your interview you will need write down at least 10 different questions to ask this person during the interview. You need to include these questions and the person's answers in your project's presentation.
3. After your research and interview you need to include a few paragraphs headed: "What I learnt about food processing ..." This must be a summary of the information you learnt.
4. Present your findings in a visible form that you like - perhaps a poster, PowerPoint slide show or in a flip-file as a brochure. Be creative and present your information in a lively and interesting way!
5. Use the knowledge and understanding that you gained and process the food using this method.
6. You will need to present your poster/presentation/slideshow to the class as well as the steps you followed to process your chosen food. Bring some of your processed food to class so that you can all enjoy what everyone has made.
7. When you choose a processing method make sure you have food that you will be able to process using this method! If you cannot find fresh (raw)

peaches to poach because it is winter, then consider choosing another processing method!

8. Please list all resources (referencing) that were used in your research. (acknowledgements)

Teacher's Note

Below is a guideline rubric to assess each learner's project:

Rubric

	5	4	3	2	1
Research skills	Rigorous pursuit of knowledge & understanding; information paraphrased very clearly.	Good coverage of information and new knowledge / understanding; some evidence of paraphrasing	Adequate and fairly precise knowledge gathered. Mostly copied information.	Information gathered by copying and pasting information in random without much planning.	Not much information gathered. Some information is space-filling and serves little purpose.
Interview skills	Questions and answers displays very clear understanding of processes involved and information needed	Questions are clear and to the point though not always focused. Some answers are well structured	At least ½ of questions are well structured / formulated but not always relevant and focused	Some questions were formulated well but these are not in any particular order and do not flow one from the other.	Questions don't always pertain to the topic. Some questions are difficult to understand. Did not formulate 10 questions.
Ability to synthesise information in a summary	Summary creates a very good overview of main ideas. Clear and	Summary is fairly concise and contains many main	Summary is in places a copy of the previous texts.	Summary is very short & contains only a few strands of	If summary is included it is no more than one or at the most two short sentences.

	precise synthesis of information from research and interview.	ideas but often supporting ideas have been included as well making it too long.	Some main & many supportive ideas included. Incomplete.	information . Imprecise and lacking	Incomplete & containing many misunderstandings
Referencing and bibliography	All references are noted and bibliography is complete and well presented.	Some sources acknowledged and referenced. Fairly good bibliography	One or two sources acknowledged. Bibliography contains titles yet incomplete.	No sources references in text but one or two sources were included in bibliography	No bibliography or referencing of sources.
General impression of presentation	Well presented, neat and creative presentation. Good use of language and resources. Interactive and lively.	Fairly well presented but could be neater. Presentation was good. Fair amount of effort applied.	Some effort applied but presentation could be far more creative and interesting. Could have used resources more effectively	Presentation was generally simplistic with little use of colour, layout or creativity. At times the reader feels confused & cannot follow	Presentation was untidy and sloppy. Information was inaccurate. Poor use of language. Very little evidence of effort applied
Effective use of information to process food	Used knowledge & understanding gained in research & interview	Fairly good understanding of food processing procedure & application in own process.	Good correlation between some aspects learnt and practical application, though not all	Applied at least 3 theoretical concepts learnt but could not sustain this without support.	Could not correlate knowledge from research with practical application.

	very effectively to plan food processing.				
Appropriate choice of food to process	Excellent choice guided by research, interview and food availability.	Considered majority of constraints & made a good choice.	Choice was made difficult as did not consider all constraints.	Chose a random food that posed tremendous problems as did not consider all constraints.	Did not prepare beforehand. Chose inappropriate food.
Outcome of the food processing	Food was very well processed and looked and tasted delicious.	Some items were processed correctly and looked and tasted fine.	One or two items were successfully processed. Others 'didn't work	Although the idea was correct the result did not work and tasted / looked unappetizing.	Lack of preparation thus food processing was unsuccessful.
Demonstration of food processing	Demonstration was lively and entertaining. Audience enjoyed it very much!	Demonstration was fairly well presented yet with a few hiccups.	Very nervous presenting yet tried hard. Audience was lost in places.	Demonstration was short / incomplete. Battled to explain the process	Tried to present / demonstrate but lacked confidence or understanding of process.
Breakdown of steps to process food	Very precise and clear breakdown of steps	Fairly precise yet not always clearly communicated breakdown of steps.	Steps were not well ordered but could break down process in main phases.	Could only identify some of the main phases but not in logical order.	Needed much prompting to identify one or two of the main phases in the process.
			TOTAL:		/50

Hint: If you have fruit that goes brown quickly, squeeze some lemon juice onto the fruit.

ADDITIVES

There are additives in some processed food. Additives are substances that are added to food products to perform certain functions such as improving taste, freshness, texture or appearance of food.

EXAMPLES OF ADDITIVES

Preservatives: preservatives, such as sulfur dioxide, are added to food to make it last longer.

Colourings: Many food and beverage manufacturers used artificial colourings to make food look more attractive to their consumers.

Artificial sweeteners: These are chemicals added to some foods and beverages to make them taste sweet. Manufacturers also used to add artificial sweeteners and flavourings, such as MS6 (monosodium glutamate) and xylitol.

Food supplements are products designed to give nutrients that might be missing from a diet. They are usually taken as tablets, capsules or powders, or as a liquid.

Some common examples of supplements include vitamins and minerals (such as vitamin C, iron and calcium), oil supplements (such as fish oil capsules) and herbal supplements.

ACTIVITY: THE PROCESS OF MAKING TRADITIONAL BEER (*UTSHWALA*)

We have learnt a lot about food processing methods in our modern lives. However, the indigenous people of southern Africa have been preserving and processing food for many generations, and still use some of their traditional methods today.

INSTRUCTIONS:

1. Read the following description of how beer is made in the Zulu culture.
2. Then answer the questions that follow.

Zulu beer making

Traditional beer (*utshwala*) forms an integral part in the Zulu culture, especially at social gatherings and traditional ceremonies. Zulu beer is traditionally made by the women. To brew the beer, the women soak coarse sorghum and maize in water for one day, typically in a big drum-like pot (*imbiza*). The following day, the broth is

boiled over a fire and more dry sorghum is added to the mixture. After this, the mixture is mashed together and then allowed to cool for the rest of the day. The next day (day 3 of the brewing process), the mixture is filtered through a sieve to remove the big fibres. The sieve is made from palm fronds and the brew is poured from the big *imbiza* vessel into the serving vessel, *iphangela*, made from clay. The beer is now ready to be served. The *iphangela* is carried from brewery in the kraal (a semi-thatched hut which allows the smoke from the fire to escape and ensures a good supply of oxygen to ferment the mixture) to the drinking assembly.

A woman scoops the beer into a drinking vessel, *ukhamba* (a small, round clay pot decorated with traditional patterns) using a dried gourd and presents it on her knees to the men. She will first taste the beer to show the head of the household that she has brewed the beer properly, and then hand him the *ukhamba*, before passing it around to the rest. The beer contains 3% alcohol and it is nutritious as it is made from plant products without modern additives or colourants.

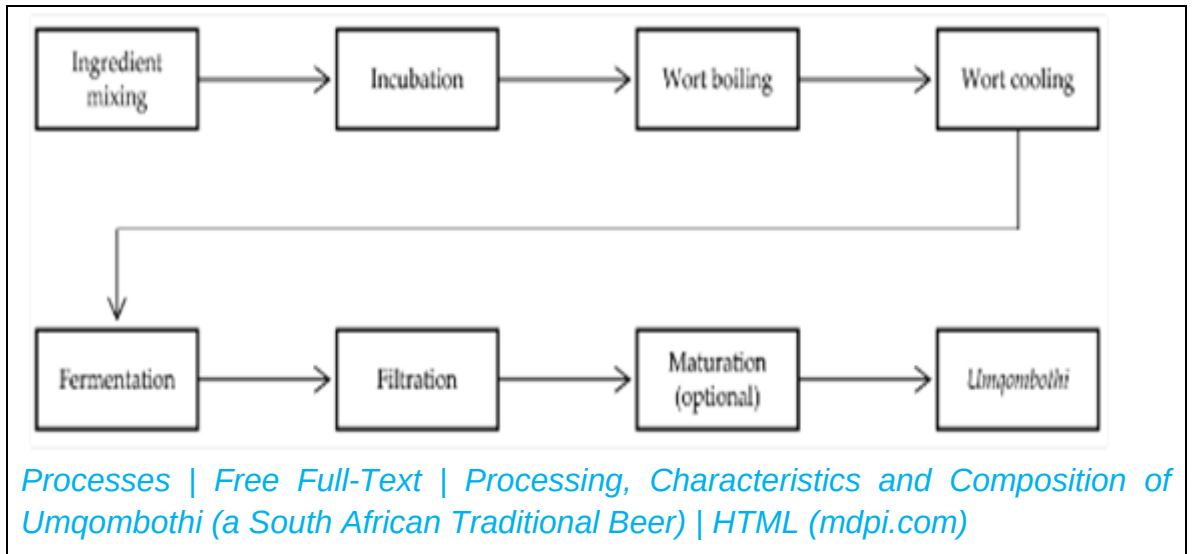


QUESTIONS:

1. What are the ingredients used to make utshwala?
Sorghum, dried sorghum, maize and water
2. What are the Zulu names for the three different clay pots (vessels) used in the beer making process?
Imbiza (brewing vessel), iphangela (serving vessel), ukhamba (drinking vessel)
3. What are these vessels made from?
Clay
4. Why does the brewery hut only have a semi-thatched roof and not a fully, enclosed roof?

This is because the women have to make a fire inside to boil the mixture. So they need an opening to allow the smoke to escape, and they need a supply of oxygen for the fire to burn and the fermentation process.

Use the space below to draw a flow diagram to illustrate the Zulu beer-making process. Remember to include arrows to show the direction.



TOPIC 4: THE DIGESTIVE SYSTEM

This chapter is intended to give an overview of the main systems in the human body, and the basic structure and function of the associated organs. The focus in each system will be to look at the, the **main components**, and then some of the **potential health issues** associated with the **system**.

The human body consist of several systems that must work together for the body to function as a whole. There are seven main organ systems in our bodies:

The Digestive system

The Circulatory system

The Respiratory system

The Musculoskeletal system

The Excretory System

The Nervous system

The Reproductive system

This chapter is an overview of one of the systems in the human body namely the Digestive system.

TEACHER'S NOTE

Our **bodies** need **protein, carbohydrates, fats, vitamins and minerals** to function. Yet we eat large pieces of food that are too big to **be taken up by our bodies**. So how does the food we eat eventually get small enough to be absorbed?

Purpose of the digestive system

TEACHER'S NOTE

Explain to learners that the digestive system is like one long tube of different **organs** and this is all coiled up inside of you. Along this tube different processes occur to digest the food we eat (namely ingestion, digestion, absorption and egestion).

Our digestive system is responsible for breaking down the food that we eat into small particles that can be absorbed into the bloodstream.

The main processes in the digestive system.

There are four main processes that occur in the digestive system.

Ingestion: Occurs when you take food into your body through your mouth by eating or drinking it.

Digestion: This is the process of breaking down large food pieces into particles that are small enough to be absorbed by the body.

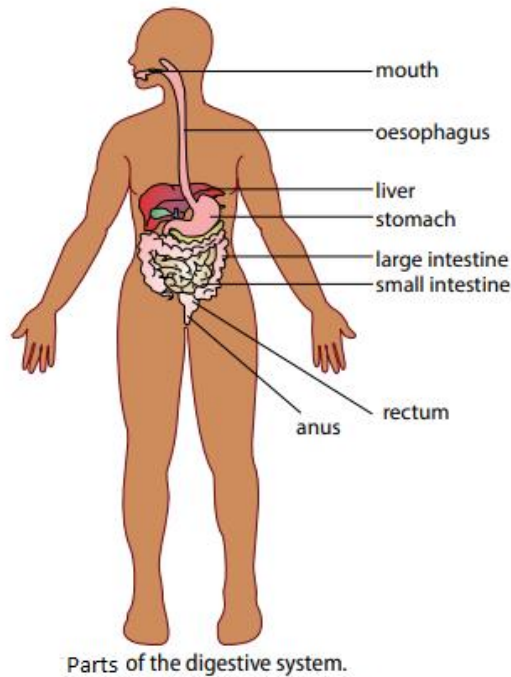
Absorption: This occurs when the digested particles move into the bloodstream from where they are carried to all the parts in the body.

Egestion: Any undigested or unwanted particles that travel through the digestive system are later passed out of the body.

PARTS OF THE DIGESTIVE SYSTEM

Structure and parts of the digestive system

Have a look at the following diagram which gives an overview of the different parts making up the digestive system.



1. The mouth and Oesophagus

Digestion starts in the mouth as food is chewed and mixed with saliva. It then travels down the oesophagus when you swallow

2. Stomach

The chewed food enters the stomach and is further digested. The stomach has substances to help digest the food. The stomach also contracts to break the food down further into a liquid.

3. Small intestine

Most of the digestion takes place in the small intestine. Absorption of the nutrients also takes place in the small intestine.

4. **Liver** the liver secretes (produces) the bile to digest fat.

5. Large intestine (or colon)

By the time the food reaches the large intestine, most of the nutrients have been absorbed. What is left is water, salts and indigestible fibre. The water that is left is absorbed in the large intestine.

6. Rectum and Anus

The remaining substances (called faeces) are passed into the rectum and then out through the anus. This is called **egestion**.

Indigestion occurs when people eat too much, too fast or food that do not agree with them. Another name for an upset stomach. Stress, not enough sleep, smoking, drinking alcohol can make indigestion worse.

ACTIVITY: Digestive System

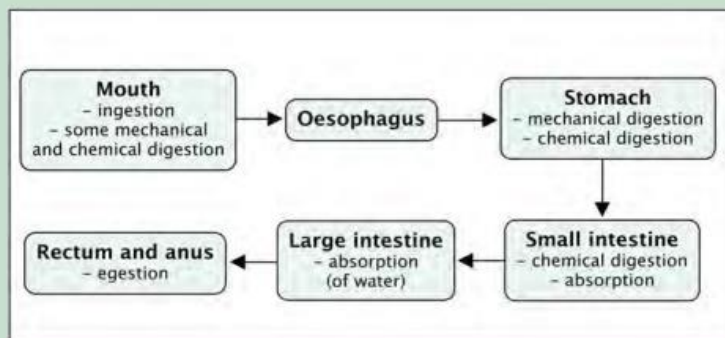
Flow diagram of the digestive system

Flow diagrams are diagrams that show how the different sets of a process fit together in a sequence. They show the direction (flow) by using arrows. These are important tools to help you think about processes in Science.

Instructions

1. Draw a flow diagram to represent the passage of food from the time it is taken into the body to the time it is egested from the body.
2. The blocks must show the main components involved in digestion, listed in order, with arrows in between. Under each component include the main processes that occur at each of these stages.

1.–2. The learners' flow diagrams could look something like this one below. An idea is to draw this up on the board and get learners to swap books with each other to check if they produced something similar and logical

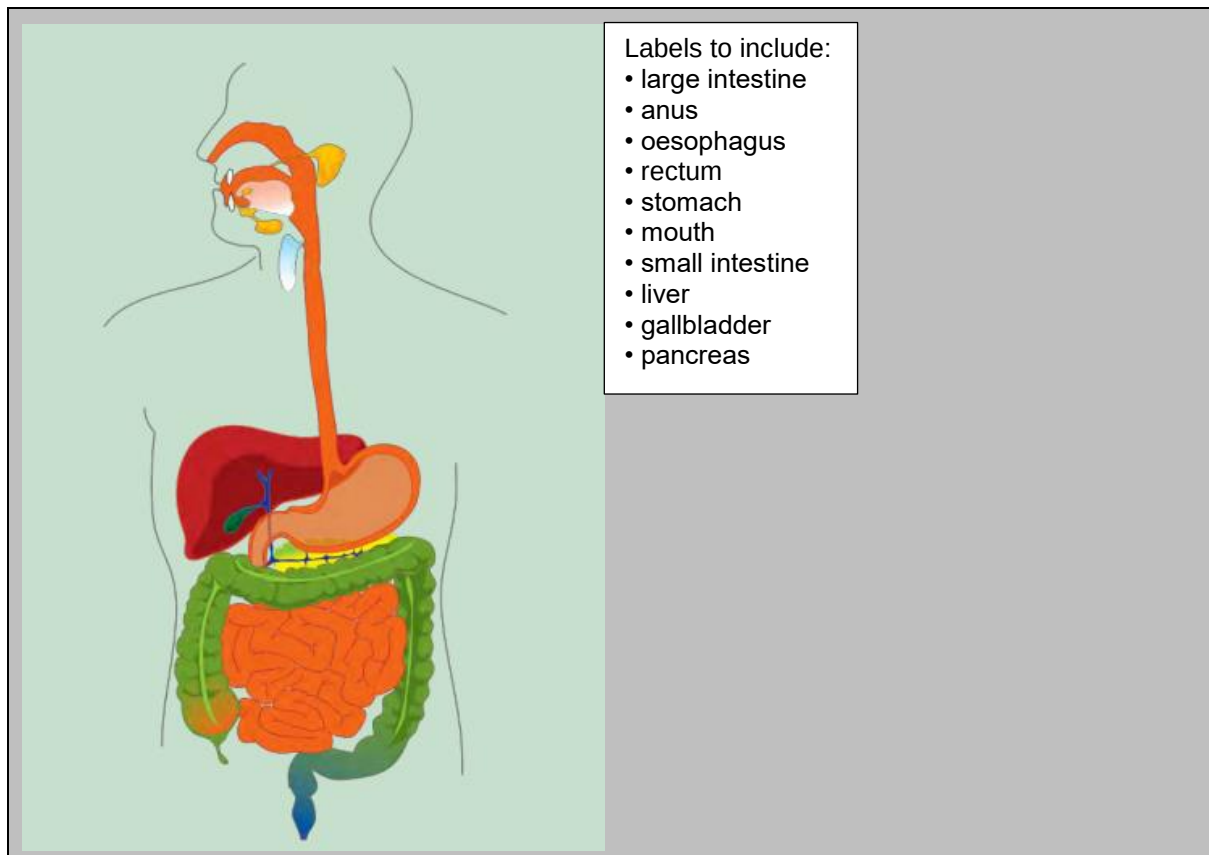


ACTIVITY: The different organs in the digestive system.

INSTRUCTIONS:

1. Label the following diagram.
2. The labels have been provided for you.

There are some which you might not have come across yet, as they are not the main components in the digestive system, but still play important roles.



TEACHER'S NOTE

Learners must draw straight, parallel label lines with a ruler and labels should be written one underneath the other.

In this image, **the liver** is the large red organ next to the stomach, and the gallbladder is the small green part in front of it. The liver produces bile which it then secretes into the gallbladder to be stored before it enters the digestive tract. Bile helps with fat digestion.

The pancreas, the yellow organ below the stomach in the diagram, is another organ which plays an important role in digestion as it produces the enzymes for chemical digestion which are secreted into the small intestine. **The liver, pancreas and gallbladder** are accessory organs to the digestive system. However, they are not part of the alimentary canal. Make learners aware that there is a difference in discussing the digestive system (including these accessory organs) and the alimentary canal (which only focuses on the organs and structures through which food passes and not the liver, pancreas and gallbladder).

Revision

1. What does digestion mean?

Breaking food into small soluble parts that can be transported in the blood and absorbed by the body

2. List the four main processes involved in the digestive system.

Ingestion, Digestion, Absorption and Egestion.

3. Describe the different components of the digestive system and their function.

Mouth and oesophagus: food is chewed in the mouth and passed through the oesophagus to the stomach

- Stomach: bolus enters the stomach where it is mechanically digested through churning and chemically digested by gastric juices in the stomach.
- Small intestine: most of the chemical digestion and absorption takes place in the small intestine where food is chemically broken into tiny pieces to be absorbed by the walls of the small intestine and transported in the blood to the cells.
- Large intestine: water that is left after digestion in the small intestines are absorbed in the large intestine and indigestible fibre is left to pass through and faeces are formed.
- Rectum and anus: undigested fibre and substances (faeces) passes from the large intestine to the rectum and anus from where it is egested

TERM 2: MATTER AND MATERIALS

CONTENT OVERVIEW TERM 2: MATTER AND MATERIALS

MATTER AND MATERIALS	<u>States of matter</u> Solids, liquids and gases - Briefly define and describe three states - Distinguish between states - Describe the properties	<u>Mixtures</u> - Briefly define concepts - Distinguish between pure substances and mixtures - Separating of mixtures (demonstrate)	<u>Acids and bases</u> - Briefly define concepts - Describe difference between acids and bases - Test various household substances with blue and red litmus paper and sort them into acids, bases and neutrals Practical activity
	<u>Metals and non-metals</u> - Define concepts - Distinguish between metals and non-metals - Properties of metals and non-metals - Uses of metals and non-metals	<u>Solutions as special mixtures</u> - Ingredients of a mixture - Explain the difference between a mixture and solution - Describe the properties of a mixture	<u>Process to purify water</u> Clean Water - Introduce and explain concepts with examples - Importance of drinking clean water - Methods of cleaning water Practical Investigation
		<u>Dissolving</u> - Explain concept - Rate of dissolving Practical investigation: Rate of dissolving	<u>Impact of weather, temperature and wind on matter and materials.</u> - Impact of weather on materials - Ways to protect materials
	<u>Processing material</u> Briefly define	<u>The water cycle</u> Briefly define concepts	

	concepts • Explain processes to make new materials or products • List the properties of raw materials used to make new products Practical: Demonstrate the changes of raw material	• Describe the stages of the water cycle • Draw the water cycle and describe in your own words Practical	
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CONTENT OUTLINE: TERM 2

WEEK	TOPIC	CONTENT
1-2	Mixtures	Mixtures of materials - Briefly explain concepts: - Mixtures - Solutions - Solid - Soluble - Insoluble - Pure Substance - Distinguish between a pure substance and a mixture - Briefly explain the properties of a mixture - Identify any four methods of separating mixtures into pure substance
3-4	Solutions as special mixture	- Name the ingredients of a mixture. - Explain the difference between a mixture and a solution.
5-6	Dissolving	- Explain the concept dissolving with examples: - List what you know about familiar situations when dissolving - Investigate the factors of dissolving Practical: Demonstrate rate of dissolving
	Water cycle	The Water Cycle - Briefly define concepts: - Water cycle - Evaporation - Condensation - Precipitation - Describe states of water on earth - Describe how matter change from one state to another - Describe the stages and features of the water cycle

		• Practical: Draw the water cycle and describe it in your own words
9 - 10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 40 % Test (Pen and Paper): 60 %		

TOPIC 1: MIXTURES

In this topic we are going to look at mixtures, their properties, methods of separation and distinguish between a mixture and pure substance.

KEY QUESTIONS

- What is a mixture?
- When is a mixture also a solution?
- How can mixtures be separated into different materials

1.1. DEFINITION OF CONCEPTS

Mixture

A mixture is two or more different materials that have been mixed together

In some mixtures, the different materials are still clearly visible after mixing.

A mixture of peanuts and raisins would be an example of such a mixture.

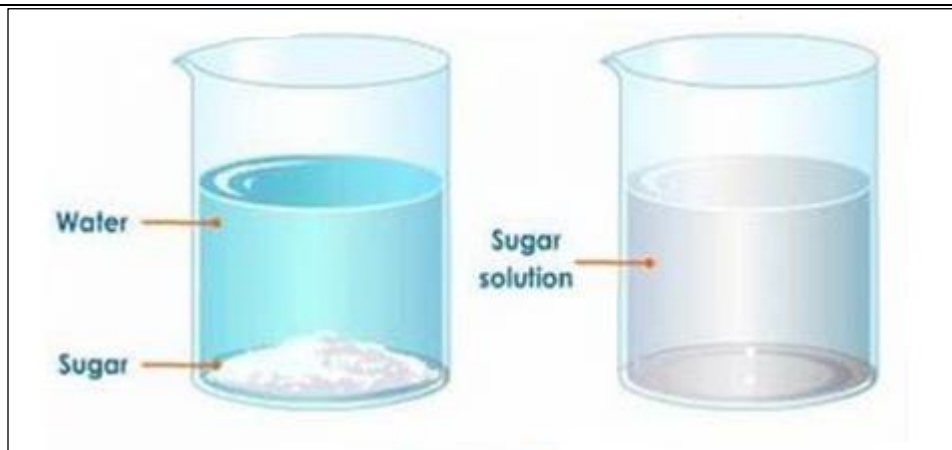


A mixture of peanuts and raisins.

Solutions

A solution is a special kind of a mixture where the components are in uniform.

In other mixtures, the materials are mixed so thoroughly that it seems one material has 'disappeared' into the other. Such mixtures are called solutions.



Solution of sugar and water

Solid

One of the 3 states of matter and it has a fixed shape. Solids can be soluble or insoluble

Soluble substances

Substances that form solutions when they are mixed with water.

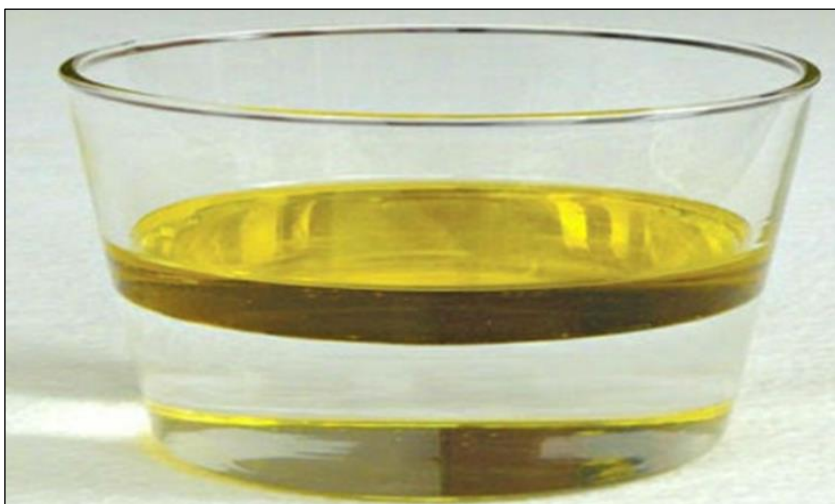
Soluble substances dissolve in water. e.g. salt and sugar are soluble in water

Insoluble substances

Substances that do **not** form solutions when they are mixed with water.

Insoluble substances do not dissolve in water.eg oil and water etc.

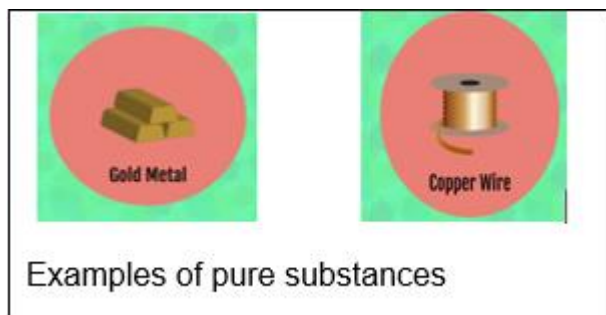
When we put them in water and try to mix them, they will not dissolve.



Water and oil (Insoluble)

Pure substances

Any material that is not a mixture.



ACTIVITY

1. What is a mixture?

Two or more different materials that have been mixed together.

2. Complete the following sentences by writing soluble or insoluble in the open spaces.

2.1. Substances that do NOT form solutions when they are mixed with water are called **insoluble** substances.

2.2. Substances that form solutions when they are mixed with water are called **soluble** substances.

3. What is a pure substance.

Any material that is not a mixture.

Now that we have learnt the concepts, let us try to distinguish between mixtures and pure substances.

1.2. DIFFERENCES BETWEEN PURE SUBSTANCES AND MIXTURES

Pure substances	Mixtures
Any material that is not a mixture	Two or more different materials that have been mixed together
Contain only one part of a substance	Contains two or more substances
Cannot be separated by chemical or physical means and its pure	It's possible to separate by physical method only and its impure

1.3. PROPERTIES OF A MIXTURE

- A mixture consists of two or more different materials mixed together.
- Sometimes mixtures can be separated into the individual materials by physical processes
- The components in a mixture may or may not be in a fixed proportion and can vary in quantity
- The composition of a mixture is not fixed.
- No chemical process is involved in the formation of a mixture.
- The properties of the components of a mixture are retained.

From the properties of mixtures, we have learnt that mixtures can be separated into individual materials by physical means, let us explore the different ways of separating mixtures.

1.4. SEPARATING MIXTURES

Some of the ways in which mixtures can be separated are: sieving, filtering, hand sorting, settling and decanting

We are going to discuss and demonstrate 4 of the separation methods

1.4.1. hand sorting and Sieving

- when the grains are large enough, we can separate them by **hand**
examples: beans and rice.
- when the two materials have grains of different sizes they can be separated by sieving. Example: sugar and rice



Sieve

ACTIVITY: Mixing and separating solids

MATERIALS:

- 500g bag of beans or lentils
- 500g bag of rice
- Small packet of sugar
- Clean sand
- Plastic spoons for scooping
- Small yoghurt tubs or paper cups for mixing
- Sieve (the type used for sieving flour)

INSTRUCTIONS: Hand sorting

1. Place 10 scoops of beans in the mixing tub.
2. Place 10 scoops of rice in the mixing tub.
3. Stir the beans and rice until they are mixed
4. Answer the questions below.

QUESTIONS:

1. Can you still see the individual rice and beans grains?
 - Yes.
2. Draw a picture of the mixture. Learners will draw in their learner's book.
3. Separate the mixture into a pile of rice grains and a pile of bean grains. Write a sentence to explain how you separated the mixture.
 - The learners may answer: 'I picked the bean grains out of the rice' or 'I picked the rice grains out of the beans'.
4. Did the beans and rice grains change in any way, or do they still look the same as before they were mixed?

- The grains still look the same

ACTIVITY: SIEVIENG

INSTRUCTIONS: SIEVING

1. Place 10 scoops of rice in the mixing tub.
2. Place 10 scoops of sugar in the mixing tub.
3. Stir the sugar and rice until they are mixed.
4. Answer the questions below.

QUESTIONS:

1. Can you still see the individual rice and sugar grains?
 - Yes.
2. Separate the mixture into a pile of rice grains and a pile of sugar grains. Write a sentence to explain how you separated the mixture.
 - The learners may answer: 'I picked the rice grains out of the sugar'.
3. Describe what you would do to separate the mixture?
 - I would put the mixture in the sieve. The sugar grains will fall through the sieve and the rice grains will stay behind.
4. Did the sugar and rice grains change in any way, or do they still look the same as before they were mixed?
 - The grains still look the same

Now that we have separated solids, let us now explore the separation of a liquid and solid

1.4.2. FILTERING

Filtering refers to separating substances by removing the solid particles from a mixture containing both solids and liquids

The collected solid material is referred to as the residue and the fluid material is the filtrate

ACTIVITY: Mixing a solid and a liquid

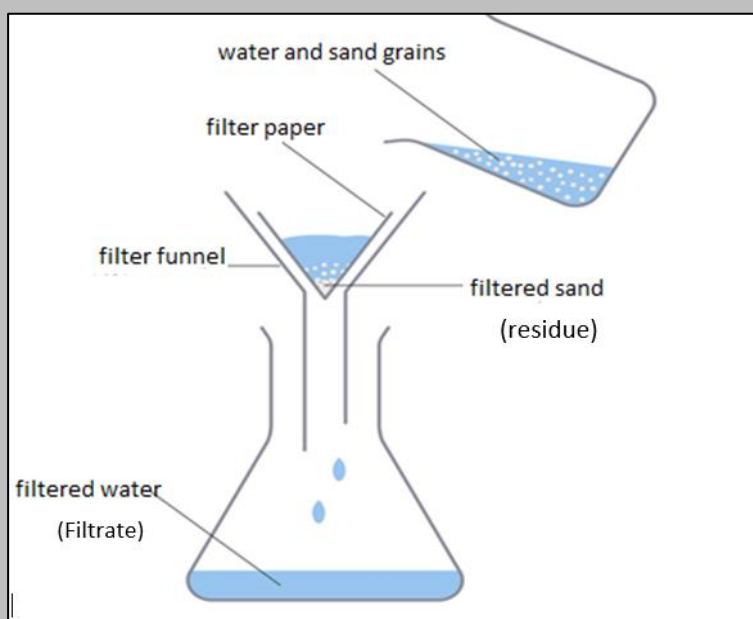
MATERIALS:

- clean sand and water

- plastic spoons for scooping
- small yoghurt tubs or a beaker
- Conical flask and a funnel
- filter paper or paper towel

INSTRUCTIONS:

1. Place 5 scoops of sand in the mixing tub. (beaker or yoghurt tubs)
2. Pour water into the mixing tub until it is half-full.
3. Stir the sand and water until they are mixed.
4. Answer the questions below:



Picture showing Filtering

QUESTIONS:

1. Are the individual sand grains still visible?
 - Yes.
2. The mixture can be separated into a pile of sand grains and water. How long would it take if you picked the sand grains out of the water one by one?
 - It would take a very long time.
3. Is it possible to separate the sand from the water using the sieve?
 - No, it is not possible, because the grains and the water would both pass through the sieve.
4. What other method can we use to separate the mixture of sand from water?
 - Settling

1.4.3 DECANTING

A mixture of two liquids can be separated by **decanting**. Example: oil and water

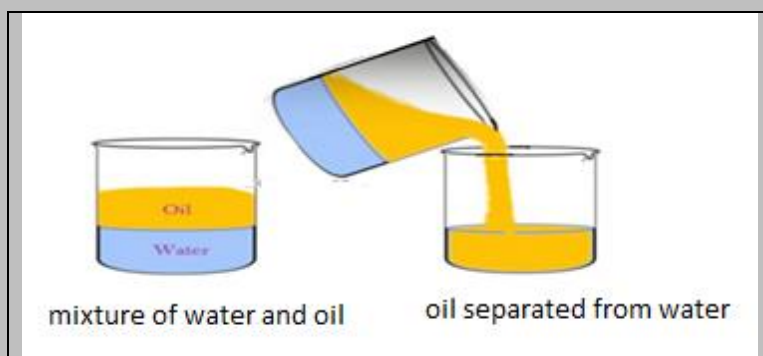
ACTIVITY: Mixing liquids

MATERIALS:

- water
- cooking oil
- plastic spoons for scooping
- small glass or transparent plastic cup for mixing

INSTRUCTIONS:

1. Place 10 scoops of water in the mixing tub.
2. Place 10 scoops of cooking oil in the mixing tub.
3. Stir the oil and water until they are mixed.
4. Let the mixture stand for a few minutes, then answer the questions below.



QUESTIONS:

1. Did the liquids mix?
 - The two liquids did not mix. There are two liquid layers.
2. Is it possible to scoop all the oil out of the water?
 - I don't think it would be possible to get all the oil out of the water'.
3. Would it be possible to separate the liquids using a sieve or a filter?
 - No, it would not be possible, because the oil and the water would both pass through the sieve and the filter.

ACTIVITY: Separating mixtures

INSTRUCTIONS

The table below contains pictures of mixtures. For each of the mixtures in the table:

1. Write which materials it is made up of in the middle column and how you would separate the mixture in the column on the right.

Picture of a mixture	Material in the mixture	How could the mixture be separated
	Peanuts and raisins	Pick the raisins out of the mixture. The peanuts will remain.
	Spaghetti and meatballs	Pick the spaghetti out of the sauce, which will remain.
	Peach, plum, banana and berries.	Pick out the berries, followed by the bananas, pieces, etc. until all the ingredients have been separated.
	Boys and girls	Tell the boys to stand on one side of the class and the girls to stand on the other side of the class.

ACTIVITY

The table below contains descriptions of mixtures.

INSTRUCTIONS

For each of the mixtures in the table:

1. Write which states it is made of in the middle column (solid, liquid or gas)
And how you would separate the mixture in the column on the right.

Description of mixture	State in the mixture	How the mixture could be separated
Water and oil	Liquid and liquid	Decant the oil off the top
Water and sand	Liquid and solid	Settling and decant , filter the mixture
Sugar and rice	Solid and solid	Pick the rice out of the sugar, or sieve the mixture through a sieve with holes that are small enough to let the sugar through but not the rice grains

SUMMARY

- A mixture consists of two or more different materials mixed together.
- Sometimes mixtures can be separated into the individual materials again.
- Some of the ways in which mixtures can be separated are: sieving, filtering, hand sorting, and decanting.
- Pure substance is any material that is not a mixture

TOPIC 2: SOLUTIONS AS A SPECIAL MIXTURE

Now that we know more about mixtures, let us explore solutions as special mixtures.

KEY QUESTION

- When is a mixture also a solution?

2.1. SOLUTIONS

Previously we looked at mixtures. We are now going to look at a special case of mixtures, which are called **Solutions**.

ACTIVITY: When is a mixture also a solution?

In this activity we are going to mix substances with water to see which ones make solutions.

MATERIALS:

- small quantities of the following substances: sugar, salt, sand, oil, vinegar, flour, copper sulphate
- tap water
- clean yoghurt tubs (small)
- plastic spoons for scooping and stirring

INSTRUCTIONS:

1. Half-fill a yoghurt cup with tap water.
2. Place one small scoop of sugar in the water and stir it well.
3. Look at the mixture and discuss what it looks like.
4. At the top of the table below, a few possible observations are given. Choose the one that is the best description for what you observed, by making a cross in the matching column.
5. Once you have recorded your observation, you can empty the yoghurt tub.

6. Repeat steps 1-4 until you have tested all the substances in the list.

Substance	Observations	
	None of the substance has dissolved	All or most of the substance has dissolved
Sugar		X
Salt		X
Sand	X	
Oil	X	
Vinegar		X
Copper sulphate		X

ACTIVITY

1. Which of the substances seemed to be uniform and the components cannot be seen when mixed with water?

- Sugar, salt, vinegar and copper sulphate

2. Which of the substances in this activity did NOT form solutions with water? (Hint: which ones did not look as if they 'disappeared' into the water?)

- Sand and oil

A solution is a mixture where the components are in uniform. We cannot see the components, however, we can see traces of some substances in solutions, example the colour of blue copper sulphate dissolved in water.

- The substance that have dissolved is **a solute**.
- The substance in which a solute is dissolved **solvent**.
- The solvent and solute together are called the **solution**

ACTIVITY:

Which mixtures are solutions?

In this activity we will use our observations from the previous activity (When is a mixture also a solution?) to decide which of the mixtures we made were solutions

QUESTIONS:

1. What do we call the substance that dissolved in water?

- Solutes

2. What do we call water in the mixture?

- solvent

3. What name do we give to these mixtures?

- Solution

4. Complete the table to tell whether the mixture is a solution or not.

Mixture	Is the mixture a solution after stirring? (Yes ?No)
Sugar and water	Yes
Salt and water	Yes
Sand and water	No
Oil and water	No
Vinegar and water	Yes
Copper sulphate and water	Yes

In the activity above, we mixed different substances with water.

1. Which substance is the solvent in all the mixtures

- water

2. From the mixtures above,

2.1. Choose an example of a solution that consists of a solid solute and a liquid solvent.

- Any one of the following examples: Sugar and water, salt and water or copper sulphate and water

2.2 Choose an example of a solution that consists of a liquid solute and a liquid solvent.

- Vinegar and water

2.3 Choose an example of a mixture of two liquids that is NOT a solution

- Oil and water

In the next activity we will look more closely at a solution, in order to understand how it is possible for the solute to dissolve in the solvent.

ACTIVITY:

What is a solution?

MATERIALS:

- copper sulphate crystal (any other household substances that will dissolve in water and change colour)
- tap water
- clear container, such as a glass beaker or test tube (a water glass will also do)
- plastic spoon for scooping and stirring

INSTRUCTIONS:

1. Look at the copper sulphate crystals and the water. Write one sentence to describe each substance in the table below.
2. Mix one small scoop of the copper sulphate with enough water to dissolve it completely (half a cup of water should be enough). Let it stand for few minutes until it clears.

Teacher's Note

Teacher's note: ensure that all of the copper sulphate has dissolved, or learners will be confused by some of the results of this activity.

1. Look at the copper sulphate solution and write a sentence to describe it in the table below. Save it for answering the questions that follow.

Substance or mixture	Description (what it looks like)
Water	The water is a colourless liquid
Copper sulphate crystals	The copper sulphate is a blue solid
Copper sulphate solution	The solution is a blue liquid

QUESTIONS:

1. What evidence do you have that there is copper sulphate in the water?

The water is the same colour as the copper sulphate crystals.

2. Can you see any copper sulphate crystals moving about in the water?

No.

4. What happened to the copper sulphate particles?

The copper sulphate particles have been mixed up with the water particles.

2.2. DIFFERENCES BETWEEN A MIXTURE AND A SOLUTION

MIXTURE	SOLUTION
- Two or more different materials that have been mixed together. - We can easily identify the substances that has formed the mixture.	- A special kind of a mixture where the components are in uniform. - We cannot see the components, however, we can see traces of some substances in solutions.

ACTIVITY

Decide whether the mixture is a **solution** or **not**.

	The mixture is a solution	The mixture is not a solution
Can you see the solute in the solvent?	No	Yes
Does the solute settle out?	No	Yes
Can the mixture be separated using filtration?	No	Yes
Can the mixture be separated by filtration?	Yes	No

ACTIVITY:

Indicate whether the substance that was mixed with water are

1. mixtures or solutions,
2. soluble or insoluble

Mixture	Is the mixture a solution? (Yes or No)	Is the substance that was mixed with water soluble or insoluble
Sugar and water	Yes	Soluble
Salt and water	Yes	Soluble
Sand and water	No	Insoluble
Oil and water	No	Insoluble
Vinegar and water	Yes	Soluble
Copper sulphate and water	Yes	Soluble

Summary

- A solution is a special kind of mixture. Like all mixtures it consists of two (or more) substances mixed together.
- A solution is made up of a solvent (such as water) in which one or more solutes have been dissolved.
- In a solution, the solute looks as if it disappears into the solvent. This is because the particles of the solute and the solvent become closely mixed.
- There are many kinds of solutions, but the most well-known ones are mixtures of a solid and a liquid, such as sugar and water.
- Not all substances dissolve in water. Those that dissolve are called soluble substances; those that do not dissolve are called insoluble substances.
- Solutions cannot be separated by sieving, filtering, hand sorting, or settling and decanting. This is because solute particles are dispersed between the solvent particles.

TOPIC 3: DISSOLVING

KEY QUESTIONS.

- What is dissolving
- What does rate of dissolving mean?

In this topic we are going to explain the concept dissolving, investigate the factors of dissolving and demonstrate the rate of dissolving.

What is dissolving?

Dissolving involve two materials separating into parts and the resulting solution is a mixture of the two materials, the particles of a solute spread throughout the particles of a solvent. we say the solute dissolves in the solvent to make a solution.

Example: sugar dissolve in water, salt dissolve in water.

3.1 RATE OF DISSOLVING

The rate of dissolving refers to how quickly a solute dissolve in a solvent. There are 3 factors that affect the rate of dissolving and we will explore them as we proceed.

ACTIVITY:

We are going to perform 3 practicals to explore the factors that affect the rate of dissolving of a salt or sugar (a solute) and Water (a solvent.)

Practical no.1

AIM: To investigate how temperature affects the rate of dissolving.

MATERIALS AND APPARATUS:

- 3 clear containers (glass beakers are ideal, but small yoghurt tubs will also be suitable)
- sugar
- very hot water (not boiling)
- tap water (at room temperature)
- ice water
- teaspoon
- stopwatch or clock with second hand

METHOD:

1. Measure the same quantity (100 ml) of tap water, hot water and ice water respectively into the three containers. Look at the diagram of the setup.
2. Place one teaspoon (5 ml) of salt into the container with the hot water.
3. Stir the solution by moving the teaspoon once back and forth through the water.

Teacher's Note

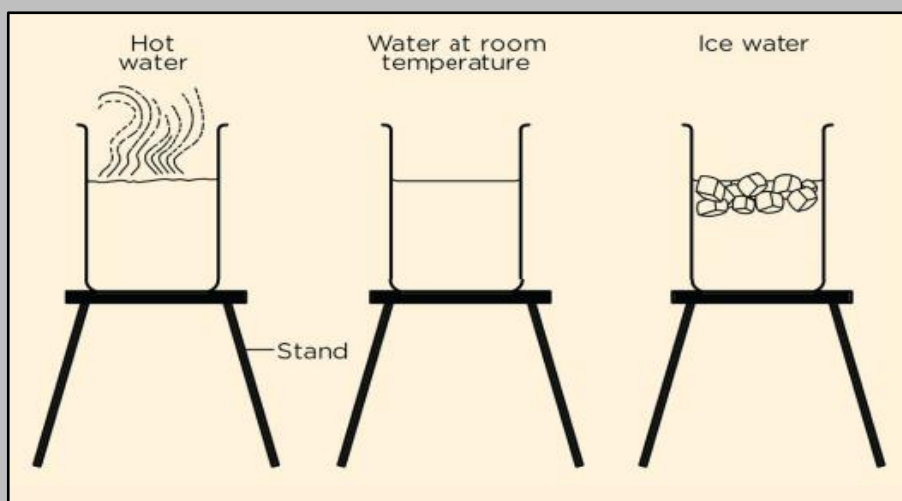
Stirring must occur at the same, regular rate for comparisons to be drawn. Let each group agree on a way to ensure this (perhaps by counting each full "stir" of the spoon)

4. Measure the time it takes for the sugar to dissolve completely. Record the time in the table below.
5. Repeat steps 3-5 with the tap water and record the time it takes for the sugar to dissolve in the table.
6. Repeat steps 3-5 with the ice water, and record the time it takes for the sugar to dissolve in the table

Teacher's Note

This step should be done with the ice OUTSIDE the water used for mixing. Either very cold water from a fridge can be used, or the mixing container should be placed on a bed of crushed ice. The presence of the ice in the mixing water obviously has nothing to do with the rate of dissolving. Additionally, some of the sugar may end

up on top of the ice and freeze there, creating a situation where it may appear that the salt dissolved quicker in the cold water than the room temperature water



RESULTS:

The effect of temperature on dissolving.

Situation	Time it takes to dissolve (in seconds)
Sugar in hot water	
Sugar in tap water	
Sugar in ice water	

QUESTIONS

1. What did we compare in this investigation?

We compared how fast sugar dissolved in hot water, water at room temperature, and ice water.

2. Name three things that were the same about the three situations.

The amount of water we used.

The amount of sugar added to the water.

The amount of stirring.

3. What did we change in this investigation?

The temperature of the water.

4. We measured the temperature of the water. What else did we measure?

We measured the time it took for the sugar to dissolve.

5. Complete the conclusion below by filling in the missing words.

The sugar dissolved **faster** in the hot water than the cold water.

Temperature affects the rate of dissolving.

When we increase the temperature of the solvent, the rate at which the solute dissolves **increases**

Practical no. 2

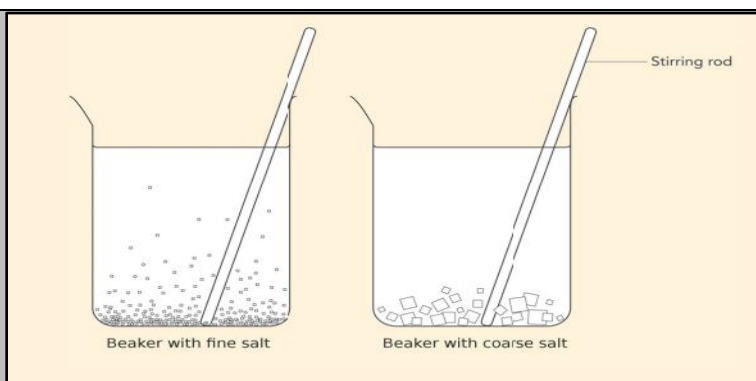
AIM: *To investigate how grain size affects the rate of dissolving.*

MATERIALS AND APPARATUS:

- 2 clear containers (glass beakers are ideal, but small yoghurt tubs will also be suitable)
- fine table salt
- coarse rock salt
- teaspoon
- tap water
- stopwatch

METHOD:

1. Measure the same quantity (100 ml) of tap water into each container.
2. Place one teaspoon (5 ml) of fine table salt into the first container.
3. Stir the solution and measure the time it takes for the salt to dissolve completely. Record the time in the table below.
4. Repeat steps 2-3 with the coarse rock salt



RESULTS:

The effect of grain size on dissolving

Situation	Time to dissolve (in seconds)
Fine salt in water	
Coarse salt in water	

ACTIVITY

1. What did we compare in this investigation?

- We compared which dissolved faster in water, fine salt or coarse salt

2. Name three things that were the same about the three situations.

- The amount of water we used.
- The temperature of the water.
- The amount of stirring.

3. What did we change in this investigation?

- The grain size of the salt.

4. What did we measure?

- We measured the time it took for each type of salt to dissolve.

5. Complete the conclusion below by filling in the missing word or words.

The fine salt dissolved **faster** than the coarse salt.

Grain size affects the rate of dissolving.

When we increase the grain size of the solute, the rate at which the solute dissolves **decreases**.

PRACTICAL NO.3

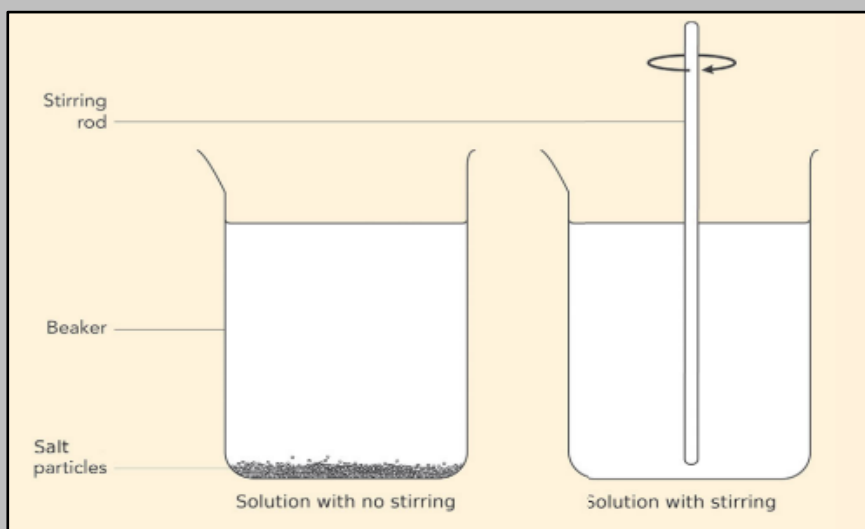
AIM: To investigate how stirring affects the rate of dissolving

MATERIALS AND APPARATUS:

- 2 clear containers (glass beakers are ideal, but small yoghurt tubs will also be suitable)
- fine table salt
- tap water
- teaspoon

METHOD:

1. Place a teaspoon of salt in each container.
2. Add the same amount of water (half a cup) to each container.
3. Stir the mixture in the first beaker but not the mixture in the second beaker.
4. Record your observations in the table



RESULTS:

The effect of stirring on dissolving

Situation	Time to dissolve (in seconds)
Salt in water (with stirring)	
Salt in water (without stirring)	

QUESTIONS:

1. What did we compare in this investigation?

- We compared which dissolved faster in water, a salt solution that is stirred or a salt solution that is not stirred.

2. Name 3 things that were the same about the 3 situations.

- The amount of water we used.
- The temperature of the water.
- The amount of salt we used.

3. What made the salt dissolve faster: stirring, or not stirring?

- Stirring

4. Complete the conclusion below by filling in the missing word or words

The mixture that was stirred dissolved **faster** than the mixture that was not stirred. Stirring affects, the rate of dissolving.

Summary

- The time it takes for a substance to dissolve is called the rate of dissolution.
- The rate at which a substance dissolves can be affected by 3 factors, namely:
 - the temperature of the solution;
 - whether or not the solution is stirred (or shaken); and
 - the grain size of the solute.
- A solute will generally dissolve faster if the solvent in which it dissolves is warm.

TOPIC 4: WATER CYCLE

4.1 Definition of concepts.

Evaporation is a process that takes place when water changes from the liquid state to the gas state.

Condensation it is a process that happens when water changes from gas state to the liquid state

Precipitation takes place when the water droplets in the clouds get bigger, some of the water falls as rain, snow, etc.

Water cycle is an ongoing process of water always changing from liquid to a gas and gas to liquid.

Liquid water evaporates into water vapour, condenses to form clouds, and precipitates back to earth.

The state of water on earth.

People say the Earth is the blue planet, because much of its surface is covered in water and the land forms a small part.



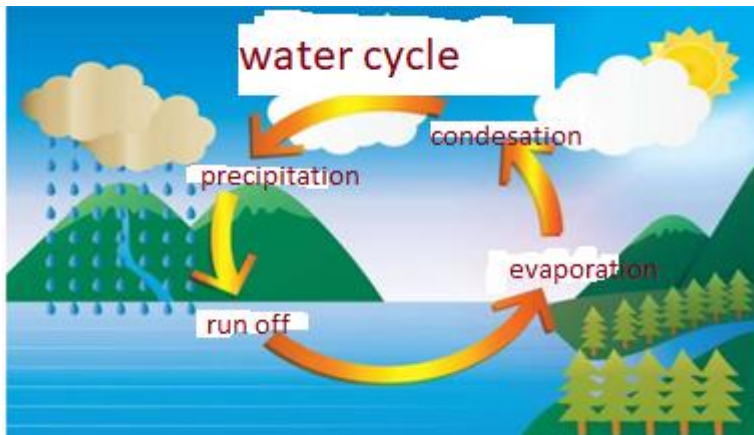
Picture of the earth

Did you know that the amount of water on Earth now is about the same as when the dinosaurs lived on our planet? How is that possible? The answer is that invisible water vapour in the air cools and condenses to form drops of water. The reverse process takes place when water evaporates. When the water evaporates, it cannot be seen anymore as it has become a gas called water vapour. This ongoing process of water always changing from a liquid to a gas and gas to liquid is called the water cycle. In a cycle, a set of events keep on repeating in the same order. Let us now take a look on the water cycle.

Stages in the water cycle

Teacher's Note

Get your learners to first discuss the picture below showing the water cycle before the facts are given.



- The Sun's heat causes water to evaporate from the seas, streams, rivers and lakes.
- The **water vapour rises into the air**.
- Higher up where the air is cooler, water vapour **condenses** into millions of water droplets which form cloud.
- When the water droplets in the clouds get bigger, some of the water falls as rain. The science word for this process is **precipitation**.
- In other clouds which become really cold, the water vapour freezes and forms snow. The snow falls down to the ground and **melts**.
- Some runoffs water that falls to the ground flows down the rivers to the seas.
- And this water will evaporate again forming part of the water cycle

Look at the image again which shows the water cycle. Use the picture to describe the water cycle.

Let's make a model to help explain the water cycle. Models are very important in science as they help to show an important process or concept in real life. A model is something we build to represent or explain what happens in real life.

TEACHER'S NOTE

- The following activity involves making a model of the water cycle

- . Models are very important in science as they help to demonstrate a concept or process. Explain to the learners that different parts of the model will represent actual things in real life. For example, the water in the bottom of the bottle will represent the ocean.
- At the end of the activity the learners will need to answer what each part of the model represents so make sure to give hints and suggestions as you are making the model.
- It would be best to do this as a group activity. Make sure that it is a hot day and that you leave the bottles in the sun for long enough that there is evaporation and condensation in the bottle.
- An idea is to leave the bottle outside whilst carrying on with the rest of the lesson.

ACTIVITY:

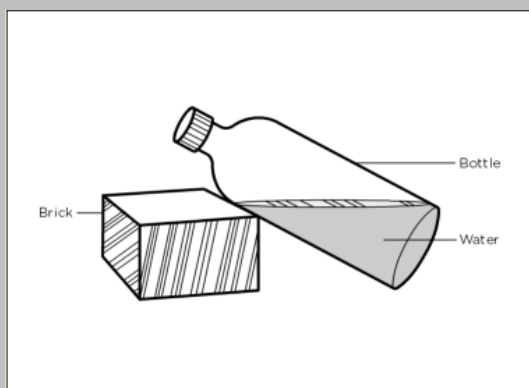
Making a model of a water cycle

MATERIALS

- a big plastic bottle (for example a 2 litre coke bottle)
- water
- a brick

INSTRUCTIONS

1. Put about a cup of water in a big plastic bottle and put the lid back on.
2. Rest the upper part of the bottle on a brick as shown in the diagram.
3. Leave the bottle in the sun for about 20 minutes.
4. Observe what happens and write down your observations.



Set up for the model of the water cycle

Write down your observations

QUESTIONS

1. Which part of the model is like the sea?

- The water in the bottom

2. Which part is like rain falling?

- The water condensing on the side of the bottle

3. Which part is like the river flowing back to the sea?

- The droplets running back down to the water along the side of the bottle.

4. What do we call the process where water turns into a water vapour (a gas)?

- Evaporation.

5. Can you see how the water in the bottle is going through a cycle? Write down the cycle below.

- Water as a liquid - evaporation (water as gas)- condensation - water as a liquid

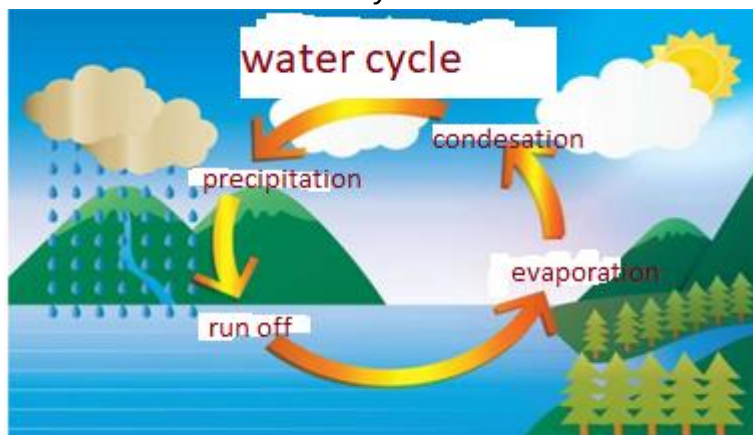
Now that we have seen a model of the water cycle, let's try drawing it.

ACTIVITY:

Drawing the water cycle

INSTRUCTIONS:

1. Draw and label the water cycle.



The water cycle

2. Use the water cycle diagram to explain in your own words how rain is formed.

Evaporation of water

Water vapour moving up

Condensation in the sky

Precipitation is rain

Learner dependent answers

[illegible]

NATURAL SCIENCES

1. CONTENT OVERVIEW TERM 3: ENERGY AND CHANGE

STRAND	YEAR 2	YEAR 3	YEAR 4
Energy and Change	<u>Energy</u> - Explain concepts with examples - Different forms/types of energy	<u>Energy and Temperature</u> - Explain the concepts with examples - Describe how to read and use a thermometer - Describe how heating and cooling can change materials	<u>Electric circuits</u> - Define concepts - Explain when a circuit is complete - Conductors and Insulators
	<u>Energy Sources</u> - Explain concepts with examples - Categorise sources of energy into Renewable and Non-Renewable sources - Name the different energy sources	<u>Stored energy in fuels</u> - Explain concepts with examples - Explain how energy is stored in food - Discuss the Renewable and Non-renewable sources	<u>Systems to solve problems</u> - Identify different circuits - Make simple electrical systems
	<u>Energy Transfer</u> - Explain where the energy from food comes from - Describe how energy is transferred in a food chain - Identify different types of energy transfer	<u>Burning Fuels</u> - Investigate, compare & describe how different fuels burn: Example candle <u>Safety with fire</u> - Identify sources - Identify safety rules when using fuels	<u>Mains electricity</u> - Describe concepts with examples - Compare the amount of energy released by different energy sources
	<u>Sound and Noise pollution</u> - Explain the concept sound with different examples - Explain sound as noise pollution and the impact on the environment	<u>Energy and Electricity</u> - Briefly define concepts - List the different types of batteries and its uses	<u>Saving Cost of Generating Electricity</u> - Explain concepts with examples - Describe energy sources and usage in South Africa - Compare different electrical appliances

ENERGY AND CHANGE CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-2	Energy and temperature	Measuring Temperature • Explain concepts with examples - Heat - Temperature - Celsius scale - Boiling point - Conductor - Insulator - Melting point - Measuring temperature Practical: • Demonstrate how to read a thermometer in degrees Celsius Demonstrate how heating and cooling can change materials
3-4	Stored energy in fuels	Fuels • Explain concepts with definitions and examples: - Fuels - Gas - Candle wax - Fossil fuels - Petrol - Paraffin - Electricity • Safety rules
5-6	Burning fuels and Safety with fire	Burning fuels • Investigate, compare and describe various fuels - Input energy needed to make them burn - Output energy released such as heat and light • Investigate compare and describe how to keep different fuels burning e.g. candle

		Safety with fire • Discuss safety measurements and safety rules when using various fuels - Practical activity: Learners should identify (for example 5) safety rules and make a poster to illustrate them.
7-8	Energy and electricity	Cells and battery • Briefly define concepts with definitions and examples: - Battery(cells) - Electric circuit - Electrical components - Stored energy - Sources of energy List and explain the different types of batteries and their uses
9 - 10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 50 % • Test (Pen and Paper): 50 %		

TOPIC 1: ENERGY AND TEMPERATURE

1.1. MEASURING TEMPERATURE

DEFINITION OF CONCEPTS:

- 1.1.1. **Heat** is the transfer of energy from one body to another as a result of a differences in temperature. When two bodies of differing temperatures come together, energy is transmitted, and heat is transferred from the hotter to the colder.

There are 3 ways in which thermal energy can be transferred from one object/substance to another, or from a system to its surroundings:

- Conduction
- Convection
- Radiation

- 1.1.2. **Temperature** is a measure of how hot or cold a substance feels, and it is measured in degrees Celsius (°C). Temperature is a measure of the average kinetic energy of the particles in an object or system



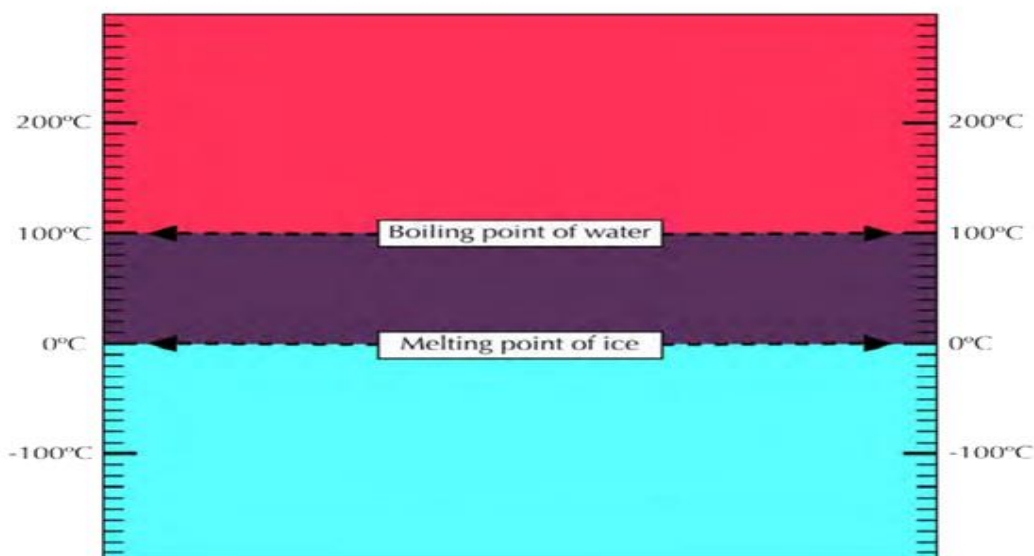
- 1.1.3. **The Celsius Scale** is one of the most widely used unit to measure the temperature.
- 1.1.4. **Boiling point** is a point in temperature at which a substance boil.
- 1.1.5. **Melting point** is a point in temperature when a solid turn to a liquid.
- Ice (the solid form of water) melts at a temperature of 0°C. This is called the melting point of ice. This is also called the freezing point of water.
 - When ice is heated, its temperature will remain constant until all the ice has melted. Only then will the temperature of the water (that was ice before it melted) rise above 0°C
- 1.1.6. **Conductor**: is a material or substance that allows heat or electricity to pass through it easily. electrical conductors are usually metal because metals generally have high conductivity.
- 1.1.7. **Insulator**: Insulators are non-conducting materials that do not easily allow current to pass through them. This does not mean that current cannot pass through them at all. For example, we generally consider air an insulator, however, lightning can cause electric current to pass through air.
- 1.1.8. **Thermometer**: an instrument for measuring temperature. It can be used to find out how hot or cold something is.

<http://youtube.com/watch?v=uLYAY-1bhBY>

BOILING AND MELTING POINT

TEACHER'S NOTE

- Some learners may know that water boils at a temperature at **100°C (100 degrees Celsius)**, depending on factors such as altitude (height above sea level) and atmospheric pressure. Do not answer this question directly, but rather use it to introduce the next activity.
- Water boils at a temperature of 100°C when the air pressure equals 1 atmosphere. This is called the normal boiling point of water. When water is heated, its temperature will rise until the water starts to boil. While the water is boiling the temperature will remain constant.



In summary:

- Boiling is a process that occurs when a liquid turns to a gas at boiling point.
- Water boils at a temperature of 100°C when the air pressure equals 1 atmosphere. this is called the normal boiling point of water. While the water is boiling the temperature remains constant.
- Melting is a process that occurs when a solid turn to a liquid.
- Ice (the solid form of water) melts at a temperature of 0°C. This is called the melting point of ice, which is also the freezing point of water.

Let's now investigate boiling and melting.

ACTIVITY 1: INVESTIGATION:

MATERIALS AND APPARATUS:

- glass beakers x 2 (or small pot)
- Bunsen burner (or stove plate)
- tripod with gauze
- tap water
- thermometer
- funnel
- ice blocks.

Before you start, discuss the following questions in your group:

1. Discuss what you know about gases, liquids and solids; the three states of matter. Write down your ideas from your discussion.
 - The particles in solids are tightly packed into fixed positions, so solids retain their shape. They cannot be compressed.
 - The particles in liquids are also close to each other, but they are not in fixed positions and can move around, that means liquids can fill the container they are in. They are not really compressible either.
 - The particles of gases are really far apart, they are not ordered in any way, they can move anywhere they want. Gases are compressible as a result. Learners should be encouraged to think about the different states of matter in terms of the particle model.
2. What needs to happen to water to make it freeze?
 - In order for water to freeze, it needs to be cooled down.
3. What needs to happen to water to make it boil?
 - In order for water to boil it needs to be heated.
4. How do we measure temperature?
 - We measure temperature with a thermometer.
5. Can you remember the boiling point and freezing/melting point of water? If you can, write them in the space below.
 - This question has been added to allow you to gauge how many learners know them.
6. Let's make some predictions. Read the two statements below, and indicate whether you AGREE, DISAGREE or are NOT SURE, by drawing a cross in the matching column:
 - Learner-dependent answers. whether you AGREE, DISAGREE or are NOT SURE, by drawing a cross in the matching column:

Statement	Agree	disagree	not sure
Water can get hotter than 100°C.			
Water always freezes at 0°C.			

Safety precautions

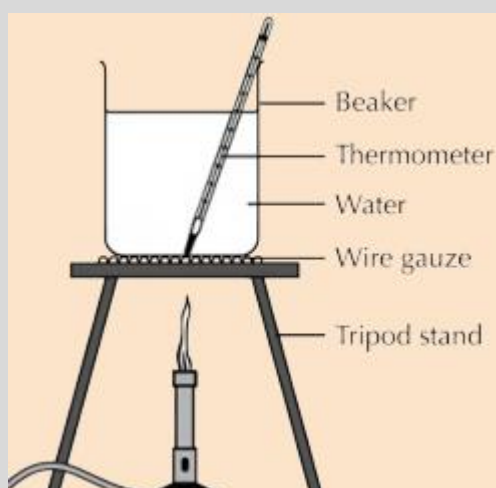
- Your teacher will demonstrate how to handle the Bunsen burner safely.
- Remember that boiling water can cause painful burns.
- The thermometer is made of very thin glass. Hold it gently, and do not use it to stir the water. Be careful not to drop it or bump it against the bottom or sides of the beaker.

TEACHER'S NOTE

- When lighting the burner, always light the match (or lighter) before opening the gas. It might be best to try this experiment in advance to determine how frequently water temperature should be measured.
- Time intervals would depend on the volume of water. CAPS suggests intervals of 3 minutes, but shorter intervals will provide more data points to plot on the graph

Set up your apparatus as shown in the image.

TEACHER'S NOTE A suggestion is to bring some ice to class and let it melt in a dish for learners to observe. Once the ice has melted, you can then leave the dish out in a hot place so that the next state change can take place and the water evaporates.



1. Remember that when you want to take the temperature, the thermometer must not be touching the sides.
2. Take a measurement of the water temperature before you start heating the water. This will be your measurement at time 0.
3. Light the burner and heat the water.

4. Measure the temperature of the water at regular intervals. Record the temperature in the table provided.
5. After a while you will notice that the temperature of the water becomes constant (this is when the temperature stops going up). Continue to take the temperature three more times (once every three minutes) after this happens. What do you notice about the water?
- 6.

TEACHER'S NOTE

Learners should observe that the water is starting to boil. When boiling starts the temperature of the water remains constant. The best graph is obtained when the water is stirred gently throughout; this ensures that the water is heated uniformly.

RESULTS AND OBSERVATIONS:

Record your results in the table.

Elapsed time (minutes)	Temperature of the water (°C)

ACTIVITY

Fill in the missing word

1. The boiling point of a material is the _____ at which the liquid state of that material turns into a gas.
The boiling point of a material is the temperature at which the liquid state of that material turns into a gas
2. When we want to measure temperature we use a _____.
When we want to measure temperature we use a thermometer.
3. The boiling point of water at sea level is. _____.
The boiling point of water is 100 °C.
4. A material that conducts heat well is said to have a high.
A material that conducts heat well is said to have a high heat conductivity.
5. If you want to create a circuit for a bulb, the material that you use in the circuit to connect the battery to the bulb must have a high
If you want to create a circuit for a bulb, the material that you use in the circuit to connect the battery to the bulb must have a high electrical conductivity

ACTIVITY 3: INVESTIGATING CONDUCTORS AND INSULATORS

TEACHER'S NOTE

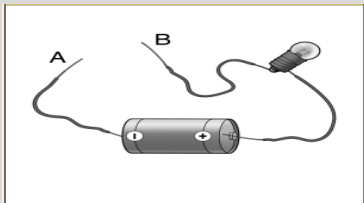
- The aim of this investigation is to show the learners how different materials affect the flow of current in a circuit
- Bring the following materials in class:
 - paper, wire, pens, rubbers, metal spoon, rubber band, coins (brass and silver) cloth, aluminium foil plastic comb, glass, steel wool, chalk, cardboard, rubber band, nail and drinking straws circuit boards/ battery, connecting wires and a bulb.
- Demonstrate with 2 materials
- Arrange learner in groups of 3 depending on the size of your class
- Distribute the materials to the groups (same materials)

Explain to learners that the flat part of the cell is negative and the one with the nob is positive

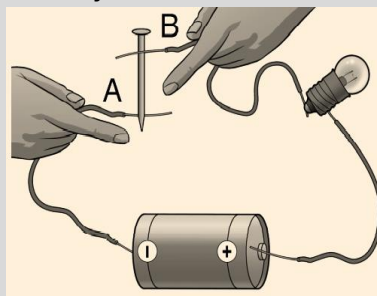
INVESTIGATION: How will you know if the material in the simple circuit conducts electricity?

METHOD:

1. Use three electrical wires to set up the electric circuit as shown below. Note that the ends of two of the wires are not touching.



2. Test that the circuit is connected properly by touching A and B to each other and making sure that the bulb lights up/
3. Take the first one of the objects in the list provided by the teacher. Put the object between the two wire ends at A and B.
4. The sketch below shows how to do it. Make sure there is good contact between the object and the wire ends.



5. Test each object as shown here with the nail
6. Does the bulb light up or not? Write the name of the object in the left or right column of the table below, depending on if the bulb lights up or not.

RESULTS AND OBSERVATIONS

Record your results below

Material	Bulb lights up	Bulb does not light up

Discussion Questions

1. What type of materials did not light up the bulb although the circuit seemed closed?
They were all made from non-metals, such as plastic, wood, glass, ceramics, etc.
2. What does that mean?
There is not an electric current in the circuit.

Note to the teacher: Summarise the investigation with the following key concepts

- Some materials allow electric current in them. They are called conductors.
- Some materials do not allow electric current in them. They are called insulators.
- Metals are usually conductors and non-metals are usually insulators.
- Electrical insulators have important functions like insulating wires or protection from electrical shock, e.g. an electrician's rubber gloves.

ACTIVITY



Look carefully at the picture. It looks as if something has gone wrong here!

1. Write a short story to explain what you think happened to the kettle in the picture. Just a few sentences are required here.

The learners should note that someone placed the plastic kettle on the stove. When the person tried to heat the water, the kettle melted as a result of contact with the flame/heat. The kettle is meant to be plugged in to heat the water.

2. Why do you think the person made the mistake of heating the kettle on the stove?
Learners can come up with their own reasons here. Perhaps the person was used to heating water on the stove in a metal kettle. Perhaps the person did not know that the plastic would melt. Perhaps the person was just absent-minded and made a mistake.
3. Do you think plastic is a good choice of material for making a whole cooking pot?
Why do you say so? No.
Plastic is not a good choice because it melts when it is heated above a certain temperature.

1.2. HOW TO READ AND USE A THERMOMETER

KEY QUESTIONS:

1. What is thermometer used for?
2. Why is the alcohol in thermometer red?
3. What alcohol is used in the thermometer?

A thermometer is an instrument for measuring Temperature Calibrated in degrees Celsius.



NOTE

- When you want to take the temperature, the thermometer must not be touching the sides.
- Take a measurement of the water temperature before you start heating the water. This will be your measurement at time 0s.

TEACHERS'S NOTE:

Explain how thermometers work

- We have different types of thermometers. In this module, we will use the commonly used one red alcohol thermometer.
- Alcohol thermometers are used to measure temperatures from -115°C to 78.5°C . The maximum temperature that can be measured depends on the boiling point of the liquid inside the bulb.
- The alcohol in the thermometer is red because since this are transparent, the liquid is made more visible by the addition of a red or blue dye.
- The red liquid in the thermometer is so thin so that it is easier to see the liquid move up and down. Since the volume of the liquid only increases or decreases a small amount when heated or cooled, it would be harder to see these changes if the tube was wider.
- We have different types of thermometers.
- In this module, we will use the commonly used one red alcohol thermometer.

Use the following link to watch the video of Thermometers



How to Read a
Thermometer _ Parts

<http://youtube.com/watch?v=uLYAY-1bhBY>

DISCUSSION QUESTIONS:

1. What do you think will happen to the liquid level in the thermometer when the liquid inside expands?
 - The liquid level will rise when the liquid expands.
2. What do you think will happen to the liquid level in the thermometer when it is put into the refrigerator?
 - It is cold inside of the refrigerator. This will make the liquid inside the thermometer contract. The liquid level will drop

1.3. HOW HEATING AND COOLING CAN CHANGE MATERIALS

NOTE

For matter to change from one state to another, its particles must gain or lose energy. The following diagram shows us that to change the state of a substance, it must either be heated or cooled.

Expansion and contraction of materials

Have you ever been inside a tin-roofed house? On a hot day, you often hear the metal roof panels groan and creak. Do you know why this happens? Some materials become slightly larger when they are heated. We say they expand. Materials can also shrink slightly when they are cooled. We say they contract. A house with a tin roof.



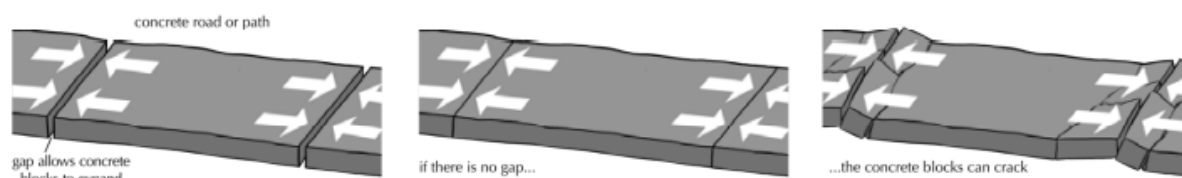
A house with a tin roof.

The metal roof panels expand and contract as the outside temperature changes. When this happens, the panels scrape against each other and against the nails that keep them in place. The scraping of metal against metal causes the creaky, groaning noises. How is it possible for materials to contract and expand? Can you think of an explanation?

TEACHER'S NOTE

Get learners to question the possibility of new atoms forming inside the material as an explanation for the phenomenon of expansion. Guide them to the law of conservation of mass: matter cannot be created or destroyed. Materials expand and contract due to particles moving further apart or closer together, not because the number of particles increases or decreases

To understand this phenomenon, we will look at some examples of expansion. We will then try to explain expansion in terms of the particle model. Some solids expand more than others. When we choose materials for a new job, it is important to know how much they will expand. This way we can allow for expansion when the materials get hot. In the following diagram, the picture on the left shows a concrete path or road surface. How have the engineers who built the road allowed for expansion?



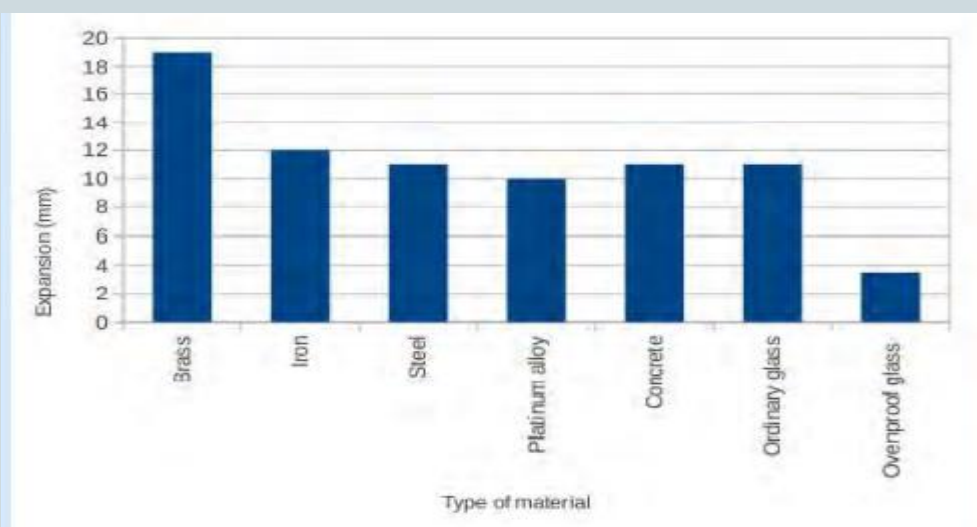
Expansion can create forces strong enough to damage materials.

ACTIVITY: How much longer

In this activity we will compare the expansion of different solid materials by drawing a graph. You will need the following information for your graph: Material.

Material	How far a 100metre length of the material will expands when the temperature increases by 10m
Brass	19 mm
Iron	12 mm
Steel	11 mm
Platinum alloy	10 mm
Concrete	11 mm
Ordinary glass	11 mm
Ovenproof glass	3.5 mm

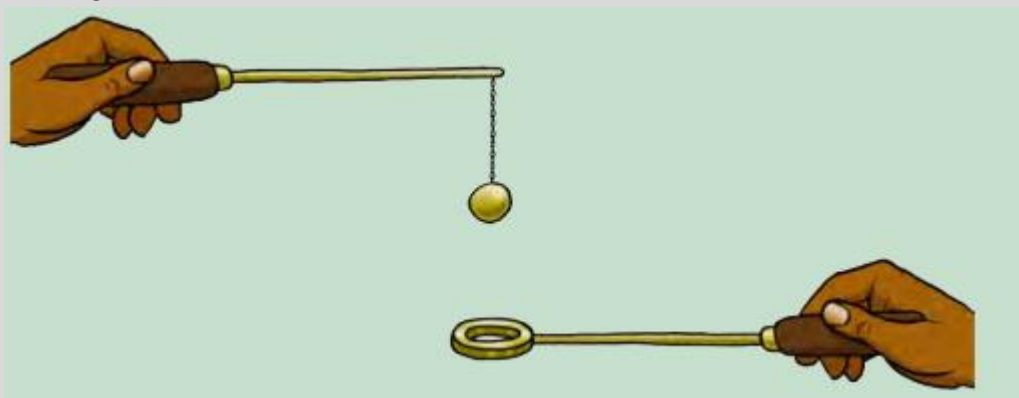
TEACHER'S NOTE Here is the bar graph that learners must draw. Any appropriate title will do, for instance: Expansion of different materials when temperature is increased by 10°C (per 100 meters of material).



QUESTIONS:

1. Which material expands the most upon heating? Brass
2. Which material expands the least?
 - Ovenproof glass
3. Which solid would be the best material to reinforce concrete?
 - (Hint: the reinforcing material should expand as much as the concrete, otherwise it will damage the concrete during expansion.) Steel. It expands the same amount as concrete.
4. The following diagram shows a metal ball and ring apparatus. The ring and ball are both made of brass. At room temperature, the ball is just the right

size to pass through the ring. Do you think the ball will still fit through the ring when the ball has been heated?



- The ball does not fit through the ring when the ball has been heated. Note: This is a great way to demonstrate contraction and expansion if you have one of these at your school. If you are doing a practical demonstration, get the learners to make a prediction before continuing with the demonstration. You can heat the ball in the flame of a Bunsen burner.
5. Do you think the brass ball will have more mass when it has expanded?
- Explain your answer. No, the brass ball cannot have a greater mass because it has not gained any mass.

TOPIC 2: STORED ENERGY IN FUELS

KEY QUESTIONS.

- What are fuels?
- What is required to burn fuels?
- How can we safely burn fuels?
- How can we prevent fires and what must we do if there is a fire?

2.1. FUELS

2.1.1. Fuels

QUESTIONS

1. What do you understand about the term fuel? Discuss this word with your partner and write down your own definition below.
 - Learner dependent answer (a material such as coal, gas, or oil that is burned to produce energy)

A fuel is a substance that is burnt and used to generate heat.

Different types of fuel include **wood, coal, oil, gasoline, and natural gas.**

2.1.1.1. Some fuels can be burnt to create heat and light

- **Wood** is often collected and burnt to give us heat and light. On a cold evening, it is wonderful to sit around a fire to tell stories and warm yourself with friends.

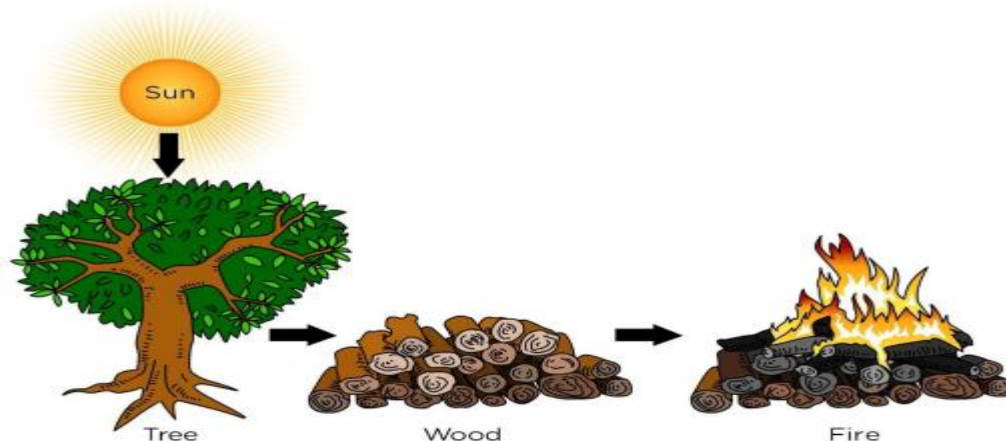


This man has made a fire to provide heat and light. ¹



Cooking meat on a fire made from wood in Khayelitsha. ²

Wood comes from plants, specifically trees. Burning wood allows us to change this stored energy into light and heat which is useful to us.



2.1.2. Gas

Gas is another fuel that can be burnt to release **stored energy** in the form of **heat and light**. We can use gas heaters to keep warm, and gas stoves to cook food and boil water.



2.1.3. Candle Wax

Wax in a candle is burnt to provide light. By burning the candle, the stored energy in the wax is released as light and heat energy.



Candle wax is an everyday fuel that we use to give us light.

2.1.4. Fossil fuels

Fossil fuels like coal, crude oil (for petrol and diesel), and natural gas were made from dead plants and animals and contains carbon. The plants took up energy from the Sun and stored it. The plants died and sank to the bottom of the water. Over millions of years, the layers of plants were covered by layers of sand and compressed by the weight of the sand. The plant material was buried deeper and deeper under the ground where it is much hotter than on the surface of the Earth. The plant remains changed into fossil fuels.

2.1.5. Petrol

Petrol are used mainly as fuel for cars, trucks and motorbikes. They are produced from crude oil, which is a fossil fuel formed from the remains of dead prehistoric animals.



*Putting petrol into a car
at a petrol station*

2.1.6. Paraffin

Paraffin is also a fuel. Paraffin is burn in lamps and paraffin stoves to provide us with useful energy in the form of light and heat.



2.1.7. Electricity

Electricity is **the flow of electrical power or charge**.

2.1.8. Safety rules

A principle or regulation governing actions, procedures or devices intended to lower the occurrence or risk of injury.

2.2. INVESTIGATE, COMPARE AND DESCRIBE VARIOUS FUELS

ACTIVITY: INVESTIGATION: How much energy can we get from different fuels?

The teacher must make the apparatus as per the instructions in the method. This investigation must be performed by the teacher or under very careful supervision due to the fire hazard. This investigation makes use of simple equipment such as tin cans and corks. But, if you have access to a science laboratory, then you can use a metal retort stand, test tube and Bunsen burner instead. The idea of this investigation is to show that you do not need fancy science apparatus to perform an experiment.

AIM: To determine which fuel contains the most energy

MATERIALS AND APPARATUS:

- a cork
- a needle
- peanuts (other fuels such as a piece of wood, candle wax or piece of biscuit)
- a large metal can (e.g. coffee tin)
- a small metal can (e.g. soup can) with paper label removed
- a can opener
- a hammer
- a large nail
- a metal spike longer than the diameter of large can
- 150 ml of water
- a thermometer
- a lighter

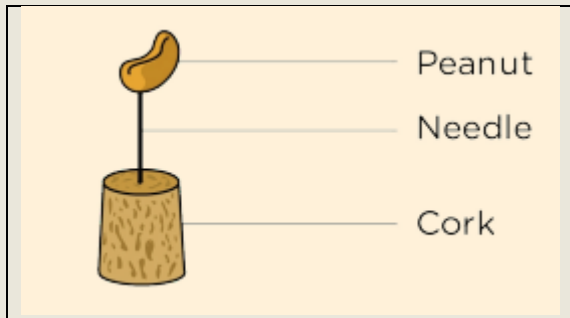
The idea of this investigation is to examine the amount of energy given off by a peanut. NB: Learners might struggle with linking a burning object, with energy given off, which then heats water, which then gives a reading on a thermometer.

Take time to explain how a burning peanut can result in a different reading on a thermometer, and that we are actually looking at the energy given off and not the reading on the thermometer. The thermometer reading is an indicator that more energy is released.

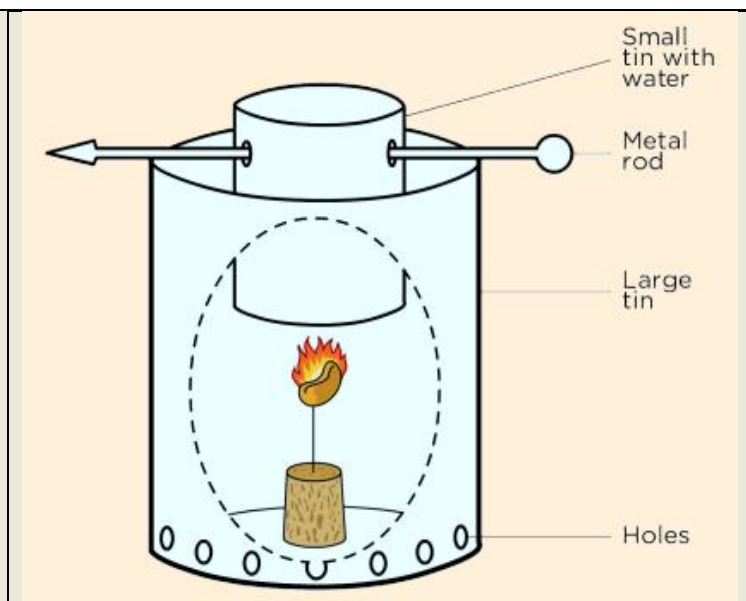
The experiment can be taken further to compare different fuels. You can also use a piece of wood, piece of candle wax, piece of biscuit (approximately the same weight of each fuel). It is important to burn the same weight of each fuel so that you can directly compare the amount of energy given off per gram of weight.

Method

1. Carefully push the eye of the needle into the smaller end of the cork. Gently push the pointed end of the needle into a peanut. If the peanut breaks use another peanut.



2. Carefully remove both ends of the large can. Watch out for sharp edges.
3. Use the hammer and nail to make holes all around the bottom of the large can. These are air holes.
4. Use the small can and punch two holes near the top of the can exactly opposite each other.
5. Slide the metal spike through the two holes in the small can.
6. Pour 150 ml of water into the small can.
7. Use the thermometer to measure the temperature of the water and record it in the results table.
8. Put the cork and peanut on a surface that cannot burn. Use the lighter to light the peanut. The peanut can be difficult to light so keep trying. It will eventually start burning.
9. As soon as the peanut is burning, carefully place the large can over the peanut. Balance the small can inside the big can as shown in the diagram. The small can must be a short distance above the peanut.



10. Let the peanut heat the small can with the water until the peanut stops burning. Stir the water and measure the temperature of the water and record it in the results table.
11. Repeat the experiment with two different fuels. Your teacher will decide which fuels to test. Fill in the results table for the other fuels tested.
 - Remember to use quantities of the other fuels which are similar in size to the peanut, and always to start with a can of cold water.

RESULTS

	Fuel 1 Peanut	Fuel 2 _____	Fuel 3 _____
Temperature of water before heating (°C)			
Temperature of water after heating (°C)			
Change in temperature (°C)			

CONCLUSION:

Write a conclusion for your investigation.

The energy stored in the peanut was changed into heat energy which we used to warm the water.

At this point, ask questions like if the peanut had stored a greater amount of energy would the final temperature of the water be greater or smaller. Lead the class to discuss which substance contained the most amount of energy.

Also ensure the learners understand that to make a fair comparison about the amount of stored energy in each substance, that you would have had to have the same mass of fuel for each experiment.

In addition, you could use a metal bottle top and fill with paraffin or other liquid fuel to compare stored energy.

QUESTIONS:

- a. Which fuel contained the largest amount of energy, and how did you determine this?
 - The fuel containing the most amount of energy would have burnt for the longest and therefore caused the greatest change in the temperature of the water.
- b. Where did the energy in the peanut originally come from?
 - The energy comes from the Sun.
- c. Discuss what happened to the energy stored in the nut, or other fuels you used.
 - As the nut was set on fire, the stored energy was released as heat and light.
- d. What was the input energy needed to make the peanut (and other fuels) burn?
 - heat energy
- e. What was the output energy obtained from the fuel?
 - heat and light energy
- f. Discuss how you could compare the amount of energy stored in peanuts to the amount of energy stored in a cashew nut.
 - The experiment (peanut or cashew nut) that produces the biggest increase in temperature has used the fuel with the most stored energy.
 - Repeat the experiment with a peanut and then a cashew nut of the same mass. Learners can go into the details of how to set up and perform the experiment. Unless they have done it, they won't be able to predict which one has the most stored energy, but the following conclusion could be made:

In order to light the fuel, you had to put in a small amount of energy. The fuel however gave out a lot more energy than what was put in. The difference between the energy you put in and the energy the fuel gave out is how much energy was stored in the fuel.

The OUTPUT ENERGY obtained from a fuel is GREATER THAN the INPUT ENERGY needed to make the fuel burn.

TOPIC 3: BURNING FUELS AND SAFETY WITH FIRE

3.1. BURNING FUELS

3.1.1. Investigate, compare and describe how to keep different fuels burning e.g. candle

We have learnt that burning fuels provides us with energy that we can use. What does a fuel need to be able to burn?

It requires some energy to start burning fuel. Fuel needs **oxygen** to burn. Fuel usually gets oxygen from the air around it. There are other gases present in air as well, but they do not burn.

When something burns we say it is combusting. Another word for **burning** is **combustion**

QUESTION

- a. How much of the air around us consists of oxygen?
 - Oxygen makes up 21% of the air around us.

3.1.2. Safety measurement and safety rules when using different fuels.

ACTIVITY: INVESTIGATION:

AIM: To find out how long a candle will burn for when given different amounts of oxygen.

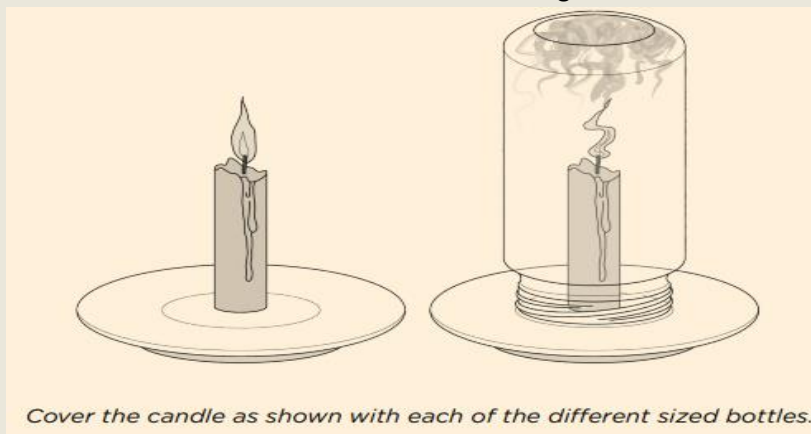
MATERIALS AND APPARATUS:

- a candle
- different sized glass bottles (small, medium, and large)
- matches
- a flat bottomed bowl
- water

METHOD:

- a. Light the candle.
- b. Drip some wax in the middle of the bowl and 'mount' a candle in the wax.
- c. Pour a small amount of water in the bowl around the candle for the glass jars to stand in.

- d. When the candle is securely standing upright, light the candle with the matches.
- e. Place the small bottle over the candle and time how long it takes until the candle goes out. Record the time taken in the results table.
- f. Repeat the experiment with each of the different sized glass containers and record the time taken for the candle to go out.



Size of glass jar	Time taken for candle to go out (s)
Small	
Medium	
Large	

1. In which glass jar did the candle burn the longest?
 - the large jar.
2. In which glass jar did the candle burn out the fastest?
 - the small jar.

NB: The outcome of this acidity can be represented in a graph.

CONCLUSION:

Write a conclusion for the investigation.

- The candle in the large jar burnt longer, because it has more air.

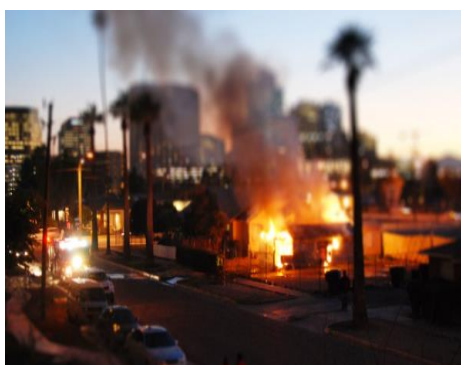
QUESTIONS:

1. When lighting a candle, identify the heat source that provides the starting energy and the fuel supply.
 - heat source is lit match; fuel is the wax.
2. Why do you think there is a difference in the time it took for the candle to go out?
 - The small jar has less air and hence less oxygen than the bigger jars. The smaller the amount of oxygen, the quicker it gets used up and the quicker the candle goes out.
3. Why did the candle go out once you put the glass jar over the candle?
 - The candle used up all the oxygen in the air. burning cannot happen without oxygen so the candle went out.

4. A candle that is allowed to burn freely in air will eventually burn down and go out. Why does the candle stop burning in this situation?
- b. The fuel (wax) has run out.

3.2. SAFETY WITH FIRE RULES

Fire is a major source of heat energy for many people whether it for keeping warm, cooking food or for some other purpose. Although fire is very useful it is also very dangerous! Great care is needed when using fire. Fire is a threat in our communities.



Invite someone from the local fire department to come and talk to the children in your school. Ask them to bring their equipment and talk to them about fire safety. The fire department is normally very willing to visit schools and they are the experts. If the fire department is not available, a community member can also be asked to talk to the learners.

SAFETY RULES WITH FIRE

1. Never play with matches and lighters. Make sure that matches and lighters are kept out of reach of young children who do not know how to use them properly.
2. In case of fire - stay away. If there is a fire in your home, do not hide, rather go outside as soon as possible.
3. Know the number of the local fire department and phone in case of emergency.
4. Have an escape plan for your home and practice it with your family. Have a meeting place outside so that you will know everyone is safe in the event of a fire.

ACTIVITY: Dangerous situations involving fire

INSTRUCTIONS:

1. Below are four different scenes.
2. Each one involves fire and is potentially dangerous.
3. Write a description next to each picture about why it is dangerous.

Situation	Why is it dangerous
	
	
	
	

ACTIVITY: Acting out what to do in case of a fire

INSTRUCTIONS:

1. In groups of 5, plan and act out a play for your class to teach them what to do in case of a fire.
2. Make sure that your play provides important information about:
 - a. how to escape from a burning building;
 - b. not to open a door in a building that is burning;
 - c. what to do if your clothes are on fire; and
 - d. what to do if your friend is stuck in a burning building.

The following actions should be shown in the plays:

- To escape from a building, fall and crawl.
- When clothes are on fire, stop, drop and roll, or cover in a blanket or carpet.
- If a friend is in a building, learners should show that they NEVER go into the building themselves, but rather call for help.

Assess the learners' plays as a group according to how clearly the learners speak and demonstrate the actions to do with their bodies and by acting out.

ACTIVITY: Creating a fire poster

MATERIALS:

Pieces of paper and cardboard.

Coloured pens and pencils.

Old magazines

Scissors

Glue

INSTRUCTIONS:

1. Design a poster telling everyone in your school what to do if there is a fire.
2. Include some pictures to show the steps to follow. You can draw these pictures or cut some out of old magazines or newspapers.
3. Some points to think about when making your poster:
 - Does your school have an alarm bell?
 - If so, what is the signal?
 - Is there a safe place that a large amount of people can gather?
 - How will you make sure no one is left inside the buildings?
 - Is it safer to use the lift or the stairs when there is a fire?
 - What extra measures can you take to stop the fire?

(Clue: Remember fire needs oxygen to burn so what can you do to your classrooms to help stop the fire and reduce the supply of oxygen?)

Extra precautions are to close windows and doors.

TOPIC 4: ENERGY AND ELECTRICITY

4.1. CELLS AND BATTERIES

KEY QUESTIONS

- What do cells and batteries do?
- What is an electric circuit?
- How can we use electricity safely?

4.1.1. Cells and Batteries

4.1.1.1. Cell

A cell is a one-piece device that turns chemical energy into electric energy. Because it just has one unit, a cell is usually small and light. A cell provides energy for a finite amount of time.



4.1.1.2. Battery

A battery is made up of a number of small parts, known as cells that store chemical energy. A battery can provide electricity for a long time. Batteries are a lot more expensive.



4.1.2. Electric circuit

An electric circuit is a pathway that allows the electric current to flow.

4.1.3. Electrical components

Electric components are parts of an electric circuit.

For example, **battery/ cell, light bulb, switch and connecting wire**. When these components are connected correctly, electricity will be transferred from one component to another.

4.1.4. Stored energy

We find stored energy in cells and batteries in the form of chemical energy.

When the battery is connected in an electrical circuit, the stored energy is transferred to the electrical circuit in the form of **electrical energy** to make it work.

4.1.5. Sources of energy

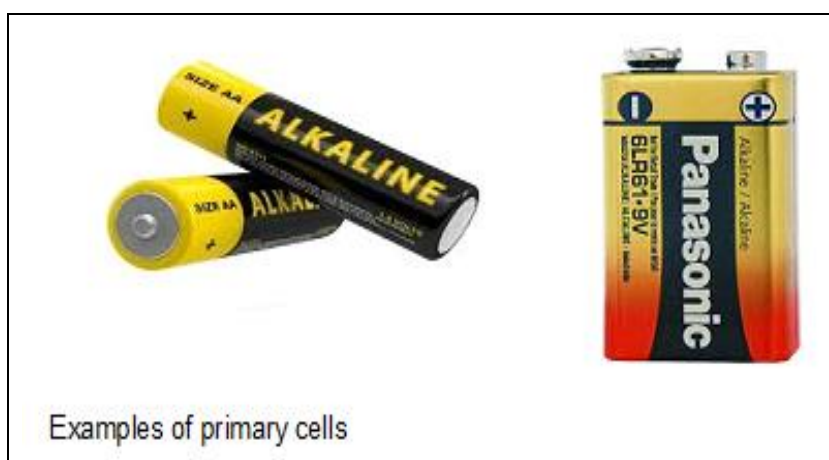
Cells and batteries are sources of energy in an electrical circuit.

4.2. Different types of batteries and their uses

Cells are categorised into primary cells and secondary cells

4.2.1. Primary cells

A primary cell cannot be recharged once it has been used.



4.2.1.1. The types of primary cells and their uses

4.2.1.1.1. Dry cells

Dry cells are used in devices such as flashlights and portable radios.

4.2.1.1.2. Alkaline cell

Alkaline cells are used in toothbrushes, toys and game controllers.

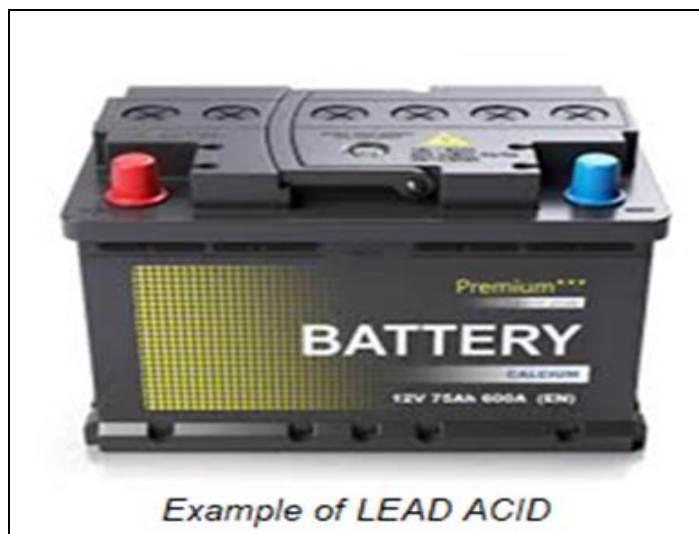
4.2.1.1.3. Leclanche cell

The Leclanche is used for telegraphy, electric bell and signalling work

4.2.2. Secondary cell and their uses

Secondary cells are the ones that are rechargeable once they have been used.

Examples of secondary cells: **Nickel-cadmium (NiCd), lead acid, and lithium ion**



They are used for:

- Starting ignition in vehicles
- Lighting in vehicles.

ACTIVITY: Batteries come in all shapes and sizes

Look at the following picture and list where can these batteries be used.



INSTRUCTIONS:

1. From the pictures above; list any three items where these batteries are used

Watch, Cellphones, Remote control, Radio and Cars

2. Look at a typical torch battery.



3. Describe in words what the battery looks like. Refer specifically to the ends of the battery. **NOTE: Allow learners to say what they see.**

Learner dependent answer.

The battery is shaped like a cylinder. One end is flat, and the other end has a knob. The battery is covered in a metallic material.

4. Look carefully to see if you can identify a positive (+) and a negative (-) sign on the battery. Which side of the battery is marked with a positive sign and which side with a negative sign?

The side with the knob is the positive end, and the flat side is the negative end.

ACTIVITY: Recycling of batteries

TEACHER'S NOTE

This activity is a good opportunity for both group work and individual work. The learners can do the research in a group but then write their paragraphs individually. Different learners in the same group may have different recycling centres closest to where they live. You can assess both the quality of their written response as well as the accuracy of their information.

Batteries which no longer work must not be thrown away in dustbins. They need to be recycled.

INSTRUCTIONS

1. Work in small groups.
2. Find out why batteries should not be thrown away in normal dustbins. Write a paragraph to explain why.
 - Batteries contain toxic chemicals which can leak into the soil and contaminate the environment. Different batteries contain different substances. Lead-acid batteries, used in motor cars and other vehicles, are particularly damaging to the environment.
3. Find out where you can recycle batteries in your community. Write down the details of the centre(s) closest to where you live.
 - This answer will depend entirely on where the learner lives. Some areas will have little to no access to specialised collection points but most Pick 'n Pay, Spar and Woolworths stores now have battery recycling collection bins and there are various companies in the country which also offer this service. Most municipal dumps also recycle batteries separately.

TERM 4: PLANET EARTH AND BEYOND

CONTENT OVERVIEW

STRAND	YEAR 2	YEAR 3	YEAR 4
Planet Earth and Beyond	<u>Planet Earth</u> - Explain concepts with examples - Describe the features of the Earth Describe the Earth as a planet in space	<u>Movements of the Earth and Planets</u> - Explain concepts with examples - Day and Night rotation The Moon	Revision of term 1 to 3
	<u>The Sun</u> Describe the unique features of the Sun Describe and compare the size of the Sun to that of the Earth	<u>Systems for looking into Space</u> - Explain how people travel into space - Describe how telescopes are used to look into space Describe the largest telescope build and used by South Africa	
	<u>The Moon</u> Describe the unique features of the moon		

CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-5	Movements of the Earth and planets	• Explain concepts and definitions: - Solar systems - Star - Earth axis - Rotation - Revolution - Satellite - Reflection of sunlight from the moon - gravity • Explain and demonstrate how the earth moves in relation to the Sun around us Day and Night • Describe the rotation of the earth on its own axis to explain day and night Moon • Explain and draw the different states of the moon • Describe the basic characteristics of the moon • Describe ways the moon states are connected to traditional, cultural activities and beliefs in the society Practical Activity: Draw a picture to illustrate the revolution of the Earth around the Sun causing the different seasons on Earth. OR: Build a model to illustrate it.
6-7	Systems looking into space	Rocket systems • Explain give definitions of the concepts: - Rocket - Telescope • Explain how people travel into space Telescopes • Describe how telescopes are used to look into space

		and gather information Discuss the largest telescope is in South Africa
8 - 10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 40 % • Test (Pen and Paper): 60 %		

TOPIC 1: PLANET EARTH AND BEYOND

1. MOVEMENTS OF THE EARTH AND PLANETS

Before we start with the content of this term

Use the following observation to record the observation of the Moon daily for 4 weeks, which will be needed later

Note to teacher: You have to give learners an activity to Observe the Phases of the Moon. Use the following table for learners to record their observations and through drawings.

The learners need to be able to describe the following general observations:

- *As the Moon orbits the Earth it appears to be changing its shape in the sky.*
- *Its shape changes from a thin crescent to a full disk (full moon) and then it shrinks back to a thin crescent again. It is then not visible for a few days again.*
- *It is important that learners also notice the different orientations of the two crescents in their respective positions.*

Observing the Phases of the Moon

MATERIALS:

- pencil to draw
- recording sheet

Instructions:

1. Look at the Moon at the same time every night, for a month.
2. Write the date in the relevant block in the recording sheet below.
3. Make a drawing of the shape you see every night in the table below.
4. See if you can identify the name of the phase and write that in the block below your drawing.
5. If you cannot see the Moon due to bad weather, then write 'bad weather' in the block instead.

Recording sheet

Week 1	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Moon shape							
Phase name							

Week 2	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
--------	-----	------	-----	-------	-----	-----	-----

Moon shape							
Phase name							

Week 3	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Moon shape							
Phase name							

Week 4	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Moon shape							
Phase name							

1.1. DEFINITION OF CONCEPTS:

1.1.1. The Solar System is the collection of eight planets and their moons in orbit around the Sun together with smaller bodies in the form of asteroids, meteoroids and comets.



The solar system consisting of the Sun and 8 planets

1.1.2. A Star is an astronomical object that produces heat and light. The nearest star to Earth is the **SUN**. Stars may appear as tiny twinkling specks in the night.

1.1.3. The Earth axis is the imaginary line that cuts through the centre of Earth from North Pole to the South Pole however it is tilted.

1.1.4. Rotation is the spinning of the Earth on its own axis. This axis is called the rotation axis.

1.1.5. Revolution is one complete movement of the Earth around the Sun.

1.1.6. Satellite is a planet, moon or machine that orbit a planet or star

1.1.7. Reflection of sunlight from the moon

1.1.8. Gravity is the tendency of all object to attract or pull each other. The pull of gravity depends on how much mass each object has and how far apart they are. The Earth is held in its orbit around the Sun by the pull of gravity.

1.2. The Rotation and Revolution of the Earth.

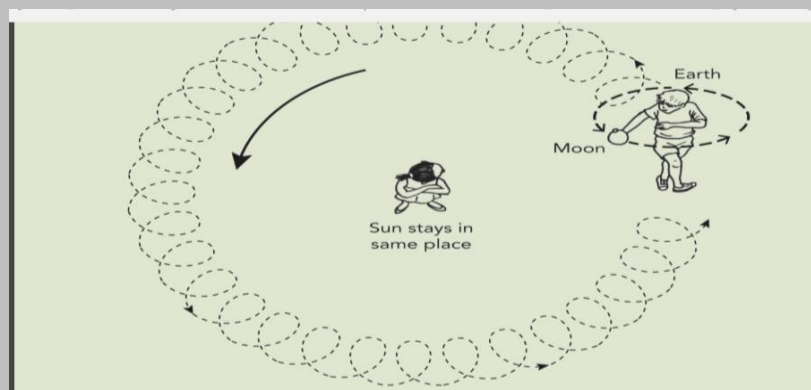
ACTIVITY: DEMONSTRATE THE MOVEMENT OF THE EARTH AROUND

Materials

- A white ball
- Two people

Instructions:

1. You must use your bodies and the white ball to act out this model of the Earth and the Moon revolving around the Sun.
2. Work together in small groups to decide how you are going to perform this model.
3. You must put together the movement of the Moon around the Earth from the last activity and the movement of the Earth with its Moon, around the Sun.
4. Perform the movements.
5. The model you must perform will be like what the following diagram shows:



The Earth revolves around the Sun in an almost perfect circle, completing one revolution (orbit) around the Sun per year (or 365 $\frac{1}{4}$ days to be precise). As the Earth revolves around the Sun it also rotates (spins) on its axis at the same time.

2. DAY AND NIGHT

During the day, it appears as though the Sun moves across the sky as it rises (comes up) in the morning and sets (goes down) in the evening.

2.1. Rotation" refers to an object's (Earth's) spinning motion about its own axis.

- The movement of the Earth on its own axis.
- Rotate means to turn, so the Earth turns around a slightly inclined imaginary line called axis
- Earth makes a full spin around its axis once every 24 hours, or one day. Earth's axis is tilted at an angle of 23.5 degrees.
- The movement of rotation causes day and night.
- During rotation the side of the Earth facing the Sun experiences daytime, and the opposite side of the Earth experiences night-time

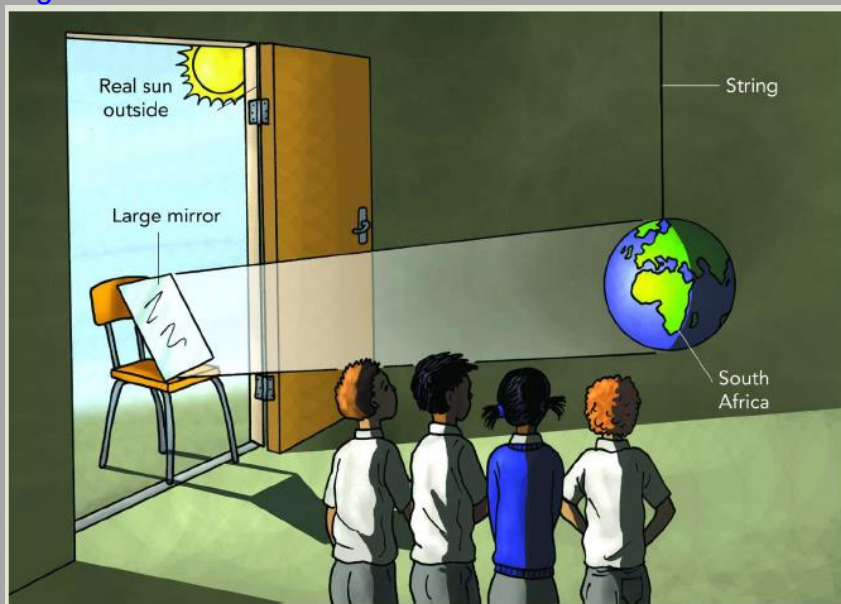
ACTIVITY: Make a model of the Earth in daytime and night time

MATERIALS:

- a globe of the Earth or a balloon with the shapes of the continents drawn on it
- string to hang a globe or balloon
- a large mirror

INSTRUCTIONS:

Teacher Note: Everyone must look at the globe from the same side.
A globe is a model of the Earth.



1. This picture shows you how to set up the equipment.
2. Set up a mirror outside the room so that it reflects bright sunlight onto the Earth globe. Your globe must be able to spin around.

3. Everyone must look at the globe from the same side.
4. Find Africa on the globe. Turn the globe so that the Sun's light falls on South Africa.
5. One side of the globe is in shadow. Find the shadow in the picture. Write the word "shadow" next to it.
This is to make sure your learners are following the explanation.
6. Find where Durban and Cape Town are on the globe.
Make small people out of Prestik and stick their feet on the globe at Durban and Cape Town.
7. Look at the globe in the picture. If you were in Durban, would it be daytime or night time?
8. If you were in Cape Town, would it be daytime or night time?
9. Now turn the globe so that Africa moves to the right. That is, you turn the globe towards the East. You will see Durban becoming dark and moving into the shadow. When Durban is in night time, Cape Town will still be in sunlight.
10. Keep on turning the globe towards the East. Now Cape Town will go into the shadow. That is night time for the people in Cape Town.
11. Which city will come back into daytime first, Cape Town or Durban?
Skill: Predicting from the pattern they can see. Answer: Durban.
We know that the Sun rises earlier in Durban than in Cape Town.
12. Keep on turning the globe to the East, and Durban will come back into daytime. You have to move to the other side of the model to see Durban move into sunlight.
13. How many hours pass for the Earth to turn around once?

2.2. WHY DO WE HAVE SEASONS?

2.2.1. "Revolution" refers to the object's (Earth's) orbital motion around another object.

- Earth revolves around the Sun,
- The Earth goes around the Sun in an elliptical orbit
- It takes the Earth 365-days (a year).

This revolution movement is what makes Spring with its colourful flowers, hot Summer full of fun, Autumn when leaves fall from the trees to the ground; and the cold frosty Winter.

TEACHER'S NOTE

A good way to introduce the concept of seasons is to have learners identify the traditional four seasons (spring, summer, autumn and winter). Ask them to describe the differences in weather and environment during the four seasons and ask them about the different activities they like to do in the different

seasons (for example going to the beach in summer). You could then ask them roughly how long each of the seasons lasts and at what time of the year the different seasons occur. In a country as large as South Africa the climate varies considerably from place to place and while it may for example be spring-like in Pretoria, it could still be frosty in the Eastern Cape. Therefore, keep in mind that the four seasons (each lasting 3 months) have been defined mainly for temperate regions and the weather expected for "spring" (mild!) might not correspond exactly to the weather experienced in a given place in South Africa at a given time. Discussions in the chapter are therefore kept as general as possible focusing on average temperatures in summer and winter for the northern and southern hemispheres.

Before investigating what causes the seasons on Earth you could ask learners what they think causes the seasons and list all the answers on a blackboard, perhaps taking a class vote for each reason. It is important that any misconceptions that learners may have regarding the cause of the seasons be addressed and some common misconceptions are explained.

You might already have some thoughts about why we get different seasons throughout the year.

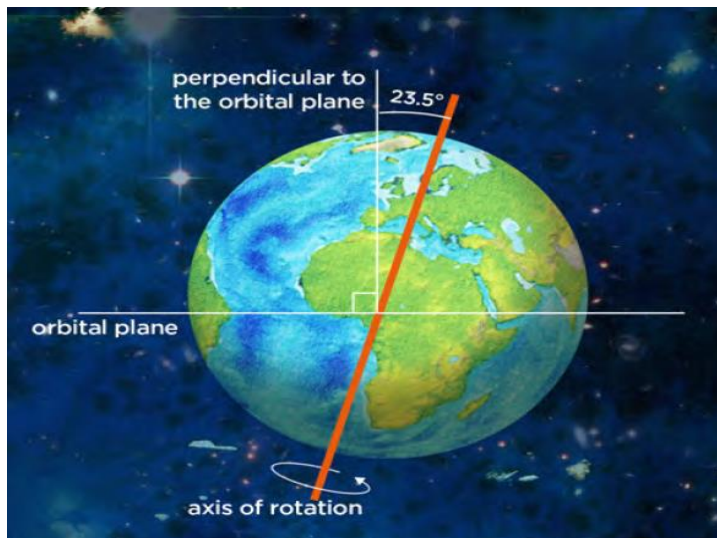
ACTIVITY: What causes the seasons? Guesses!

INSTRUCTIONS:

1. Which of the statements in the table do you think are true and which do you think are false? Put your answers in the right hand column.

Statement	True or False
We experience winter because the Sun emits less energy in winter.	False
We experience summer because we are closer to the Sun during summer.	False
If it is winter in the Northern Hemisphere it is winter in the Southern Hemisphere too.	False
Daytime is longer in the summer because the Earth spins more slowly in the summer months.	False

The Earth's rotation axis is tilted over by an angle of 23.5 degrees (23.5°) from the vertical. As the Earth travels around the Sun its North and South Poles constantly point in the same direction in space.



The Earth's rotation axis is tilted by 23.5° from the vertical as it orbits the Sun

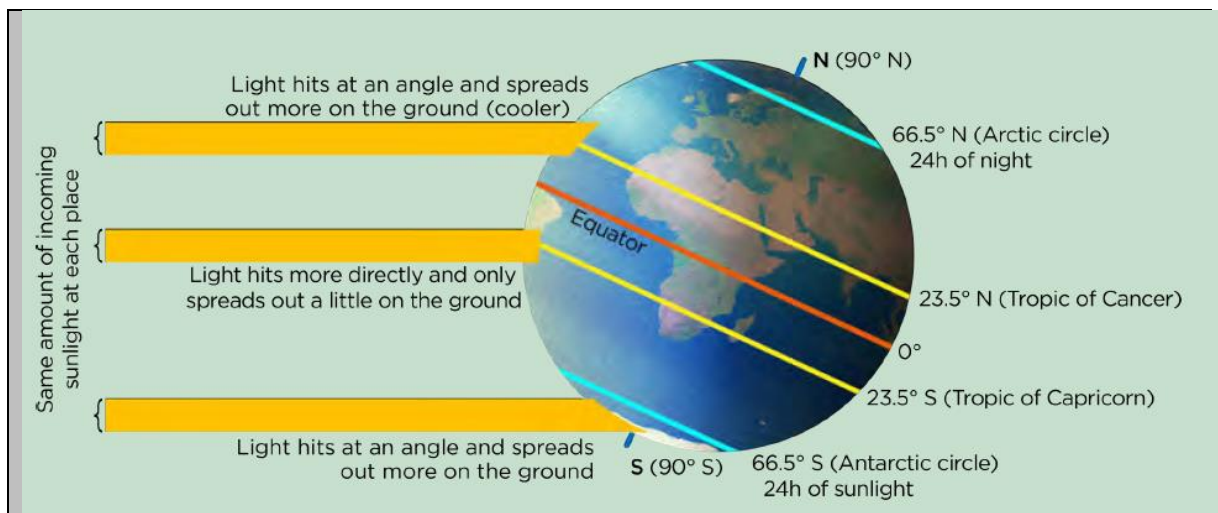
ACTIVITY: Looking at sunlight hitting the Earth

TEACHER'S NOTE

The aim of this exercise is to introduce learners to the concept that sunlight hits the Earth at varying angles across the Earth's surface because it is curved. At the equator you can see that sunlight hits the Earth almost straight on, this is called direct light. Areas close to the equator are warm as the Sun's energy is concentrated in these regions. As the Earth is curved, not all of the Sun's rays hit it directly. Areas that are hit by indirect sunlight are cooler because the Sun's energy is spread out over a large area. The poles are always hit by indirect light which explains why it is cold at the North and South Poles.

INSTRUCTIONS:

1. Look at the example picture below. It shows sunlight hitting the Earth.
2. Look at the Sun's rays and see how the angle at which they hit the Earth's surface changes at different points along the surface of the Earth because of its curved shape.
3. Answer the questions below.



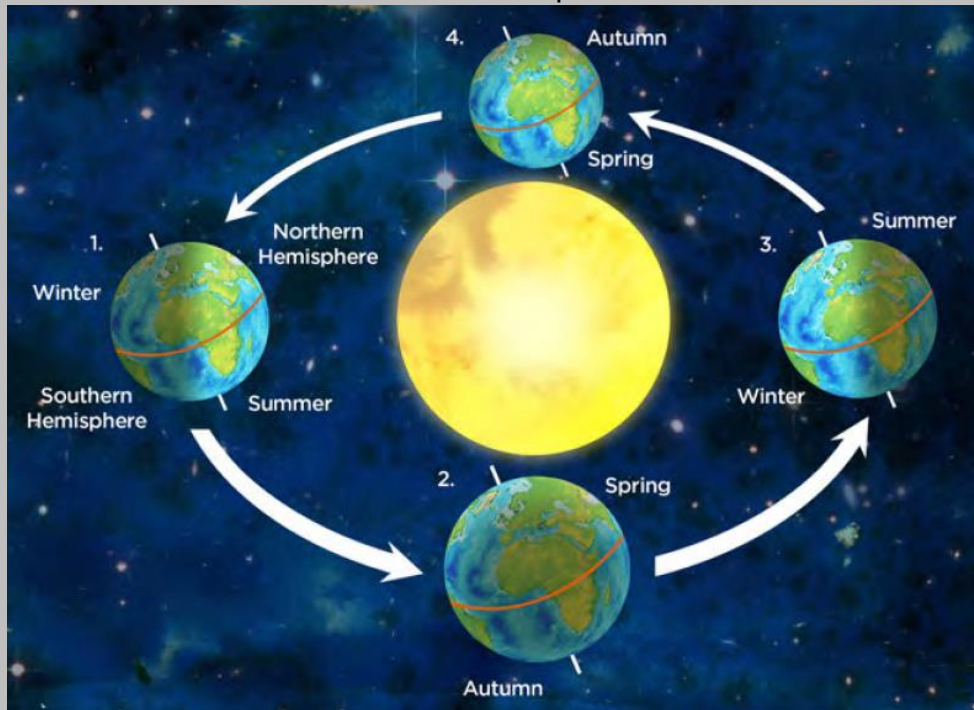
QUESTIONS:

- Does the equator receive more or less direct light than the poles?
 - More direct light.
- Which hemisphere receives more direct light in the picture? Why is this?
 - The Southern Hemisphere receives more direct light as it is tilted towards the Sun.
- Which hemisphere in this diagram receives more indirect light? Why is this?
 - The Northern Hemisphere receives more indirect light as it is tilted away from the Sun.
- Why do you think it is warmer at the equator than at the poles?
 - Because the equator receives more direct light where the solar energy is more concentrated and the poles receive only indirect light where solar energy is more spread out.
- Is it summer or winter in the Southern Hemisphere in this example?
 - Summer.
- Is it summer or winter in the Northern Hemisphere in this example?
 - Winter.
- What would happen to the seasons if the Earth was tilted in the opposite direction, with the Northern Hemisphere tilted towards the Sun instead?
 - The seasons would be reversed, it would be summer in the Northern Hemisphere and winter in the Southern Hemisphere.

. PRACTICAL ACTIVITY

INSTRUCTIONS

In this activity you will make a nice arts and craft activity to reinforce the idea that the tilt of the Earth's rotation axis is responsible for the seasons.



The seasons as the Earth revolves around the Sun.

1. Make a poster of the above figure, labelling the Earth and Sun, the equator, hemispheres, poles, tilt (towards, away, neither) and resulting seasons for each hemisphere at each position.

3. THE MOON



Note to teacher: At this point the you should already have the learners' work on the activity [Observing the Phases of the Moon](#). It is advisable that you take out the

learners' work and ask them to present their observations as represented in their drawings.

The learners need to be able to describe the following general observations:

- *As the Moon orbits the Earth it appears to be changing its shape in the sky.*
- *Its shape changes from a thin crescent to a full disk (full moon) and then it shrinks back to a thin crescent again. It is then not visible for a few days again.*
- *It is important that learners also notice the different orientations of the two crescents in their respective positions.*

Each group of learners is likely to have nearly 30 drawings of moon shapes. The teacher will help the learners to reduce these to only eight drawings to focus on major shapes, which are called phases. Only after the learners have redrawn the eight shapes (phases) of the Moon will the teacher give the learners the picture of the *Phases of the Moon* below:

3.1. Phases of the Moon

The Moon has eight phases. The Sun shines on the Moon but there is always a part of the Moon that the Sun cannot reach. We can see the part that has sunlight on it, but we cannot see the dark part that is in shadow.

ACTIVITY: Observing the Phases of the Moon

MATERIALS:

- pencil to draw
- recording sheet

Instructions:

6. Look at the Moon at the same time every night, for a month.
7. Write the date in the relevant block in the recording sheet below.
8. Make a drawing of the shape you see every night in the table below.
9. See if you can identify the name of the phase and write that in the block below your drawing.
10. If you cannot see the Moon due to bad weather, then write 'bad weather' in the block instead.

Recording sheet

Week 1	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Moon shape							
Phase name							

Week 2	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Moon shape							
Phase name							

Week 3	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Moon shape							
Phase name							

Week 4	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Moon shape							
Phase name							

ACTIVITY: Investigate the Moon

INSTRUCTIONS:

You must find information out about the Moon.

Answer the questions below.

QUESTIONS:

1. How far is the Moon from the Earth?
2. Does the Moon have air for you to breathe?
3. Is there air between the Earth and the Moon?
4. Could a bird fly from the Earth to the Moon? Give a reason for your answer.
5. Can a big aeroplane fly to the Moon? Give a reason for your answer.



Phases of the Moon

ACTIVITY: Make a model of the Earth, Sun and Moon

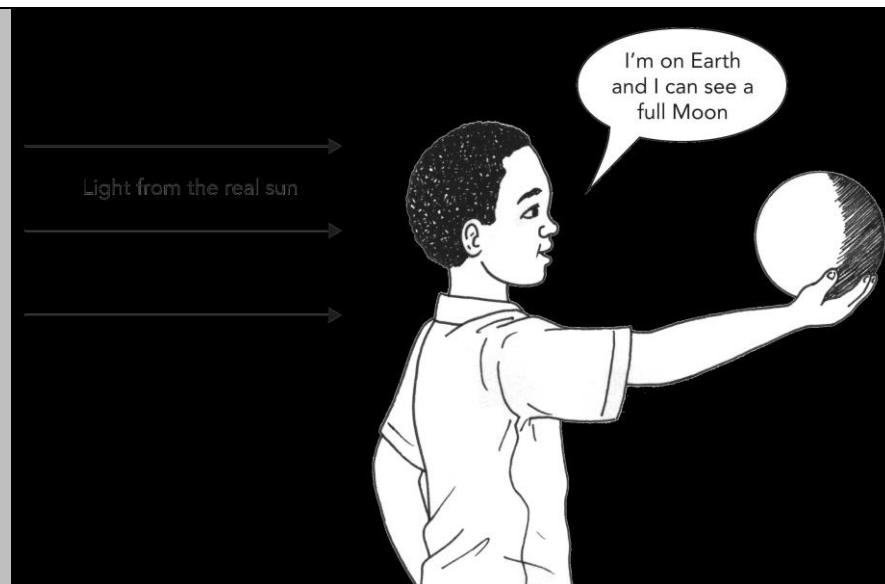
This model will help you understand why the Moon's shape changes during the month.

MATERIALS:

- a small ball to represent the Moon
- a sunny day!

INSTRUCTIONS:

1. You must do this activity outside, early in the morning while the Sun is still low.
2. Begin with your back to the Sun.
3. Hold out your 'Moon' in front of you, as you see in the following picture.
4. Your head is the Earth and your nose is Africa. You are looking from Africa.
Which phase of the model Moon do you see?



How to hold the 'Moon' with the Sun behind you.

5. Keep your arm stretched out and swing around until the model Moon is between you and the Sun. Now you see only the shadow side of your 'Moon'. You are seeing new moon.
6. Keep your arm stretched out and move the 'Moon' to your right until a little sunlight shines on the side of the 'Moon'.
7. Which shape of the Moon do you have now? Point to the pictures in the diagram showing the Moon phases.
8. Turn more to your right until half the Moon has light on it.
9. Which shape are you seeing now? Point to the picture in the diagram.
10. Turn your back to the Sun, so that you see light all over the side of the Moon. This is like 14 days passing.
11. Which shape of the Moon do you have now? Point to the pictures in the diagram.
12. Turn further to your right. Show your teacher how you can get the shape in the diagram. This is like 21 days passing.

QUESTIONS:

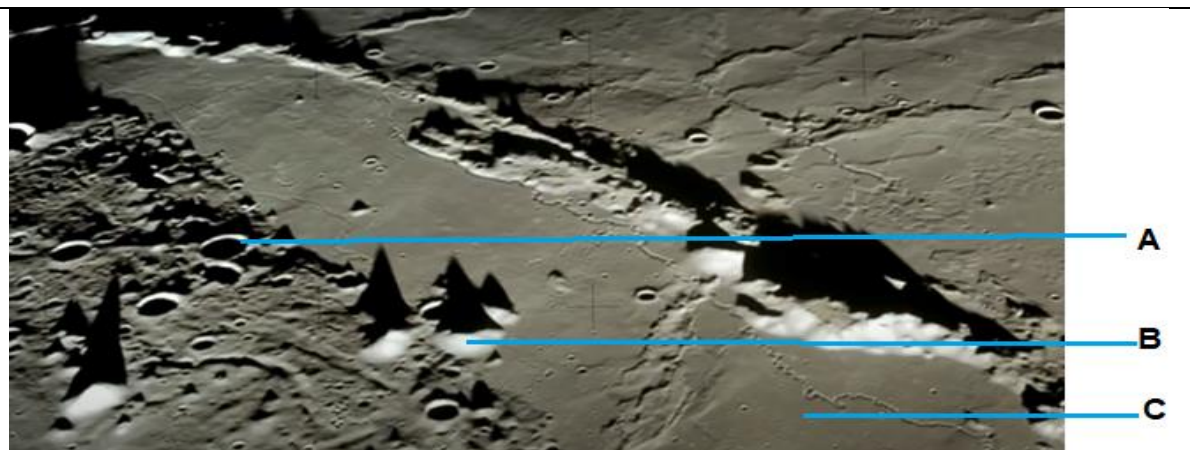
1. If it were new moon tonight, how many days will it take for the Moon to be full again?
2. How many days will it take for the Moon to be a new moon again?

3.2. Basic characteristics of the moon

Our Moon is like a desert with plains, mountains, and valleys. It also has many craters, holes created when space rocks hit the surface at a high speed. **There is no air to breathe on the Moon.**

ACTIVITY

Use the following about the Moon and answer questions that follow:



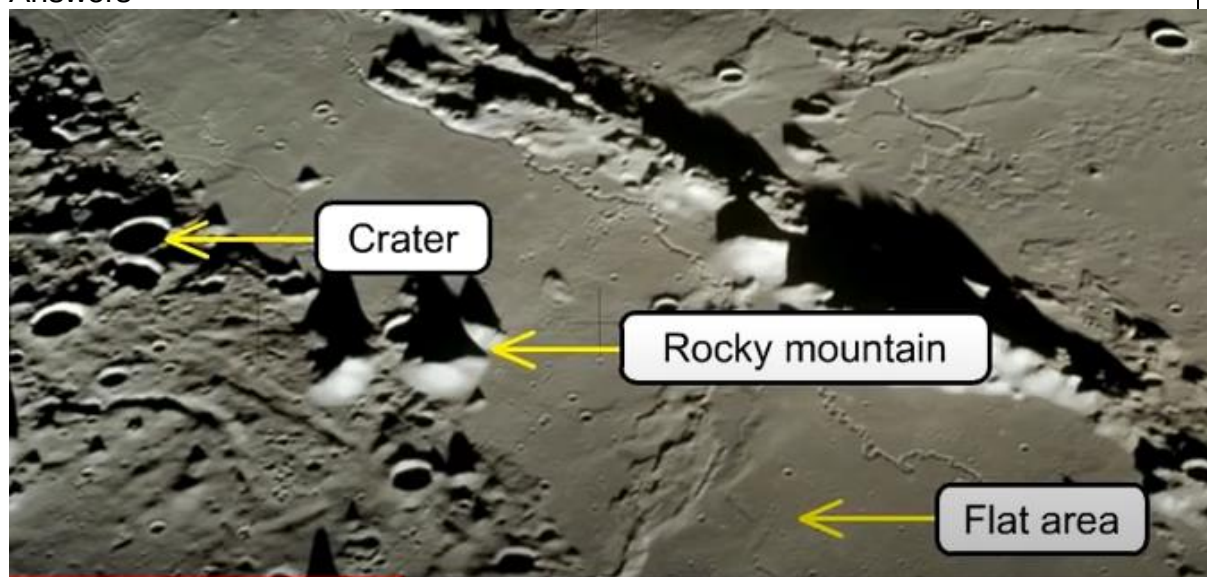
a. Identify and give meaning of your understanding of the features marked A, B, C

A _____

B _____

C _____

Answers



3.3. Moon stories

Many cultures have different stories about the Moon. These stories tell us about the importance of the Moon in people's lives.

Here are some stories about the Moon from different cultures.

3.3.1. The Moon and the Hare

The hare and the Moon met at a waterhole one night. The hare washed his face with water. The water became still, like a mirror. He saw himself in the water and then he saw that the Moon was more beautiful than he was.

So the hare took mud from the side of the pool and he threw it on the Moon's face. You can still see the mud on the face of the Moon, if you look at the Moon tonight.

QUESTIONS

- 3.3.2. What do you think the people who told this story were looking at on the Moon when they described the 'mud' on the Moon's face?
- They might have been describing craters which appear darker.
- 3.3.3. Do you think it was right that the hare threw mud at the Moon?
- 3.3.4. What emotion was the hare experiencing?
- Jealousy

TOPIC 2: SYSTEMS LOOKING INTO SPACE

Key Questions

- How can people travel in Space?
- How do I make my rocket go faster?
- How can I make my rocket go straight?

A **rocket** is a vehicle that uses a **rocket engine**. It may be a **missile**, **spacecraft**, **aircraft**.

QUESTIONS

1. Write down two things you know about rockets.
- They go up into the sky.
 - They move themselves.
 - They go fast.
 - Some rockets carry people.
 - Some rockets have gone to the Moon.
 - Gas or smoke comes out of the back of a rocket

2.1. How do rockets work?

Some things need to be found out before a rocket could be designed. In this unit, you will investigate rockets. In technology, a designer must find out what people have already made, and find out how those things work. We use the word 'investigate' which means 'to find out'.

2.1.1. INVESTIGATE ROCKET SYSTEMS

Once we leave the Earth's upper atmosphere, there is no air between there and the Moon. The wings of an aeroplane work only when they move through air.

QUESTIONS

1. Can an aeroplane fly to the Moon? Give a reason for your answer.

No, aeroplanes are not designed to fly in a vacuum (where there is no air), so their wings won't work to keep the plane in flight in Space

Rockets therefore cannot move in space the same way that aeroplanes move through the air on Earth. So rockets need to move in other ways. Let's have a look at making a simple model of a rocket to see how they move.

Teacher's Note

The next activity introduces the learners to the idea of the rocket using gas to propel forwards. The gas combusts and releases from the bottom, moving the foil in the other direction, much in the same way that a rocket does when launching and travelling in space.

Explain this to the learners during the activity.

This activity should also be **handled with caution** as matches are involved and learners could **NOT** be left alone without any supervision!!

ACTIVITY: Make a rocket from a match

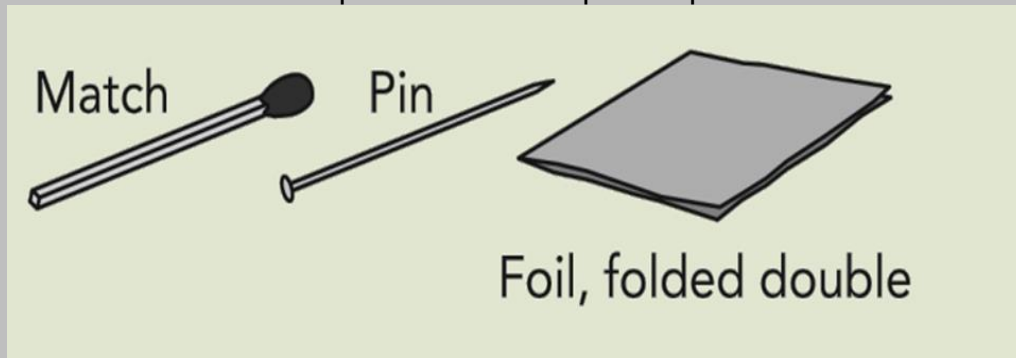
MATERIALS:

- a box of matches
- four rectangles of aluminium foil, 4 cm by 6 cm
- a pin
- a paperclip

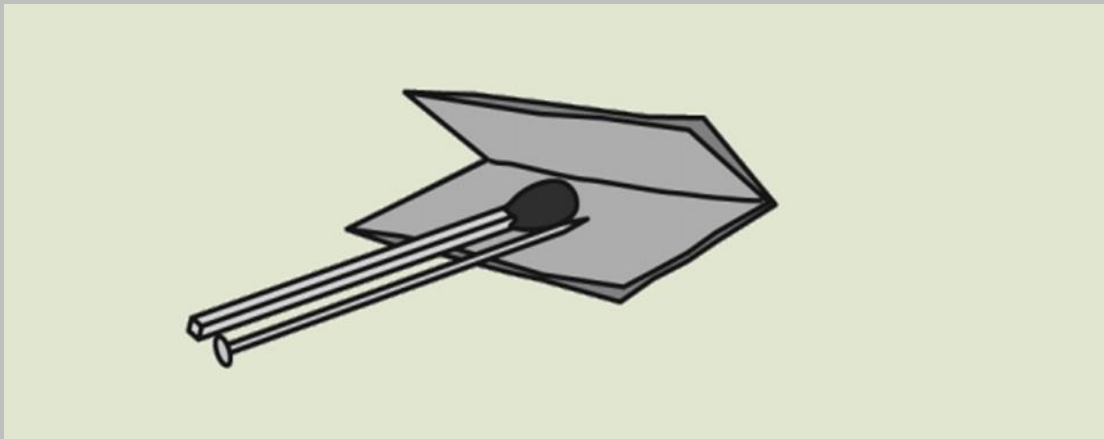
INSTRUCTIONS:

WARNING: Working with fire is very dangerous. Remove all flammable substances. Do not stand in front of your rocket during launching.

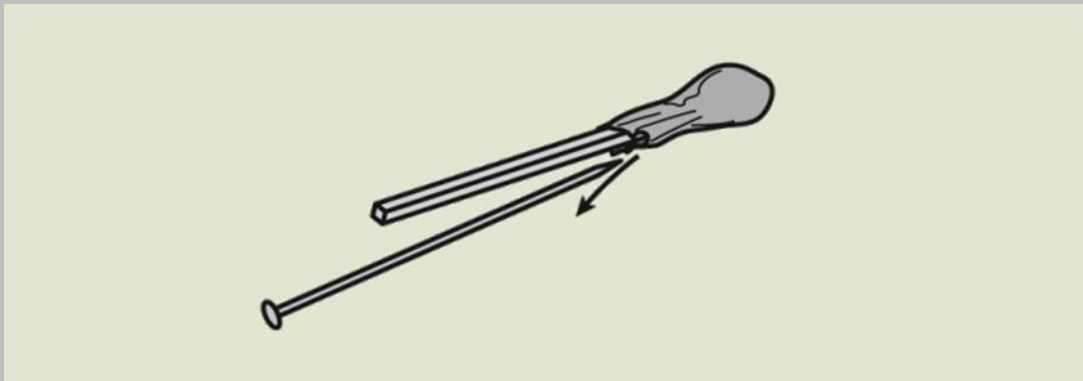
1. Put the match on the squares of foil and put the pin next to the match.



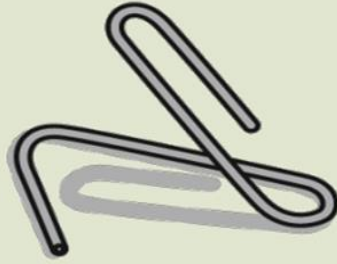
2. Wrap the foil around the pin and head of the match.



3. Pull out the pin. This leaves a tiny tube for gases to escape.



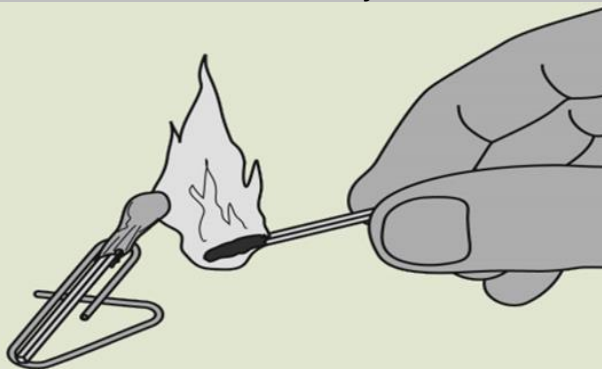
4. Now you have a rocket. It is almost ready to launch. All you still need is a launch structure.
5. Bend the paperclip to make a structure as you see in picture.



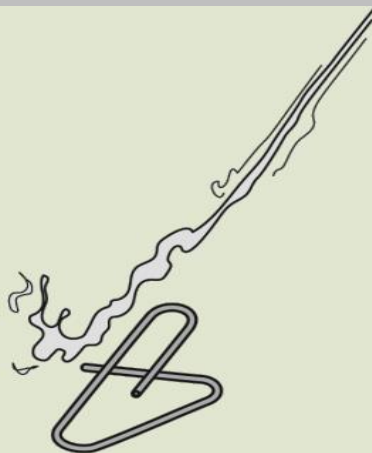
6. Put your match rocket in the launch structure. Make sure the rocket points away from people.



7. Strike the other match and heat the head of your rocket



8. Watch what happens!



QUESTIONS:

1. A rocket needs fuel. Fuel stores energy. Remember we learnt about storing energy in Term 3? Where is the energy stored in this rocket?

- **The energy is stored in the head of the match.**

So, what have we learnt from this small model of a rocket using a matchstick? When the match-head burns, it gives off hot gases. The hot gases try to expand upwards, sideways and downwards.

QUESTIONS

Make a drawing of the rocket going up. Add these labels to your drawing: nose of the rocket, tail of the rocket, nozzle, hot gases coming out.

The rocket goes upwards because the hot gases shoot downwards out of the nozzle. The faster the gases go downwards, the faster the rocket goes upwards. The size of the nozzle can affect how fast the gases come out, and that will affect how fast the rocket is pushed upward. If the nozzle is too big, the gases escape too easily. If the nozzle is too small; the gases cannot get out fast. So what is the best size for the nozzle? You can try different nozzle sizes for the matchstick rocket to find out which one works the best.

REVISION:

1. We cannot fly to the Moon in an aeroplane. Give a reason why we cannot.
 - **In Space there is no air. Aeroplane wings work only in air.**
2. Explain how a rocket moves.
 - **Rockets work by pushing hot gases out of the nozzle in the back end. This propels them forward.**
3. Give a reason why the nozzle at the back of the rocket must be small.
 - **The gases must come out very fast.**
4. What year did man first land on the Moon using a rocket?
 - **Man first landed on the Moon in the year 1968.**

2. TELESCOPES

A **telescope** makes faraway things look bigger and closer. Most telescopes, and all large telescopes, work by using curved **mirrors** to gather and focus light from the night sky.

The telescope was first discovered by the Dutch. In Holland, a lot of the people were sailors and sea-explorers so they used their telescopes at sea to see if ships far away on the horizon were friends or maybe pirates.

Galileo Galilei was a professor of mathematics at the University of Padua, Italy. In 1609 he heard that somebody in Holland had made a telescope, and he worked out how to make one himself.



Galileo Galilei's telescope looked like this. It could only see a small part of the sky at a time.¹



Galileo Galilei showing his telescope to a group of scientists

Galileo used his telescope to look at the planets in the night sky, and he made careful notes of what he saw there. He was the first person to see that Jupiter had moons. He saw that Saturn has rings and he saw that Venus has phases like the Moon has. He also used his telescope to show people that the Milky Way was really made of billions of stars. He wrote books that taught people about telescopes and what they could show us in the night sky.



People can now buy telescopes like this to watch the stars from their own backyards.²



This photo of the sky was taken through a telescope. Each point of light is a star

Note to the teacher

If you are based in Cape Town, you can take your learners to visit the South African Astronomical Observatory and if you are based in Johannesburg you can go on a tour of the Johannesburg Observatory.

Telescopes have shown us that there are thousands of millions of stars that we could not see with our naked eyes. Some of those stars are so far away that their light has been travelling for millions of years to reach us.

2.1. The Southern African Large Telescope

One of the biggest telescopes in the world is here in South Africa, near the town of Sutherland. The telescope is called the **Southern African Large Telescope (SALT)**. The telescope uses lenses and a very big mirror to see the stars and take photographs of them.



This is the SALT. Here the roof is closed, but at night it opens so that the telescope can see the sky.



This photo is taken from inside the dome of SALT and shows the large mirror.

REVISION:

1. What does a telescope do?
 - It makes faraway objects look closer.
2. Before Galileo, nobody knew that the planet Jupiter had moons. What was the reason?
 - The moons were too small to see with the naked eye, but Galileo's telescope made them big enough to see.
3. What does SALT stand for?
 - Southern African Large Telescope
4. With a telescope we can see very, very many stars. Why did we not know about all those stars before there were telescopes?
 - Those stars are too faint to see with the naked eye.
5. Astronomers build their telescopes far away from cities. Think of a reason why they do this and write it down.
 - The bright lights of the city make the sky bright and so it is harder to see the stars